For reproduction commands and a complete mapping from thesis results to the executable analysis pipelines, see Appendix **??** (especially Appendix **??**, Appendix **??**, and Table **??**).

1 Generator Model Selection

Before establishing baseline performance, we compare three frontier LLMs for CLD edge generation: GPT-4.1 (OpenAI), Sonar-Pro (Perplexity), and Claude Sonnet 4 (Anthropic). All models were evaluated on the same three validation CLDs using identical prompts (Nitai_C, T=0.7), with three independent runs per CLD.

Table 1: Generator Model Comparison: Edge Recovery Performance. Bold indicates highest F1 per CLD; in aggregate row, bold indicates highest Precision, Recall, and F1 separately. **Edges processed** refers to the number of ordered (source, target) edge slots evaluated (including `NONE`).

Model	CLD	Edges processed	TP	FP	FN	Precision	Recall	F1
GPT-4.1	Social Norms	270	17	24	16	0.415	0.515	**0.459**
GPT-4.1	Depressive	546	80	106	22	0.430	0.784	**0.556**
GPT-4.1	Emergency Dept	3168	19	356	80	0.051	0.192	0.080
Sonar-Pro	Social Norms	268	14	34	19	0.292	0.424	0.346
Sonar-Pro	Depressive	540	71	162	31	0.305	0.696	0.424
Sonar-Pro	Emergency Dept	3095	65	1184	34	0.052	0.657	**0.096**
Claude-Sonnet-4	Social Norms	270	30	74	3	0.288	0.909	0.438
Claude-Sonnet-4	Depressive	546	96	329	6	0.226	0.941	0.364
Claude-Sonnet-4	Emergency Dept	3168	92	1970	7	0.045	0.929	0.085
Micro-averaged (pooled TP/FP/FN across all CLDs):								
GPT-4.1	Total	3984	116	486	118	**0.193**	0.496	**0.278**
Sonar-Pro	Total	3903	150	1380	84	0.098	0.641	0.170
Claude-Sonnet-4	Total	3984	218	2373	16	0.084	**0.932**	0.154

Note. All models used identical prompts (Nitai_C variant) and temperature (T=0.7). Results pool TP/FP/FN counts across 3 independent runs per CLD (same configuration; stochastic sampling). Nodes (variables) were directly retrieved from the **GT lit CLDs**. **Edges processed** denotes the number of ordered (source, target) edge slots evaluated (including `NONE`), i.e., ideally $|V|(|V|-1)$; it can be slightly lower if some rows are missing due to parsing/processing errors.

Table 1 shows the edge recovery performance of the three generator models. GPT-4.1 achieves the best aggregate F1 (0.278) through balanced precision-recall. Sonar-Pro performs competitively and wins the Emergency Dept CLD (F1=0.096), but its micro-average aggregate F1 is still lower than GPT-4.1 due to substantially more false positives overall. Claude-Sonnet-4 shows highest recall (0.932) but lowest precision (0.084), generating many false positives and yielding a lower aggregate F1. Based on best micro-average F1 score, **GPT-4.1 was selected** as the generator model for all subsequent experiments. Additionally, GPT-4.1 is the only model that provides token-level log probabilities (logprobs), enabling confidence-based edge filtering which is necessary for RQ2 experiments.

2 Generator Performance Baseline

Having selected GPT-4.1 based on the model comparison above, we now establish its baseline performance. These results show that the CLD generator recovers substantially more ground-truth edges than a structural random baseline (under our GT Lit operationalization) using the Nitai_C prompt at $T = 0.7$.

2.1 Random Baseline Comparison

To confirm the generator produces meaningful causal structure rather than random edge configurations, we compare LLM-generated edges against a Monte Carlo random baseline.

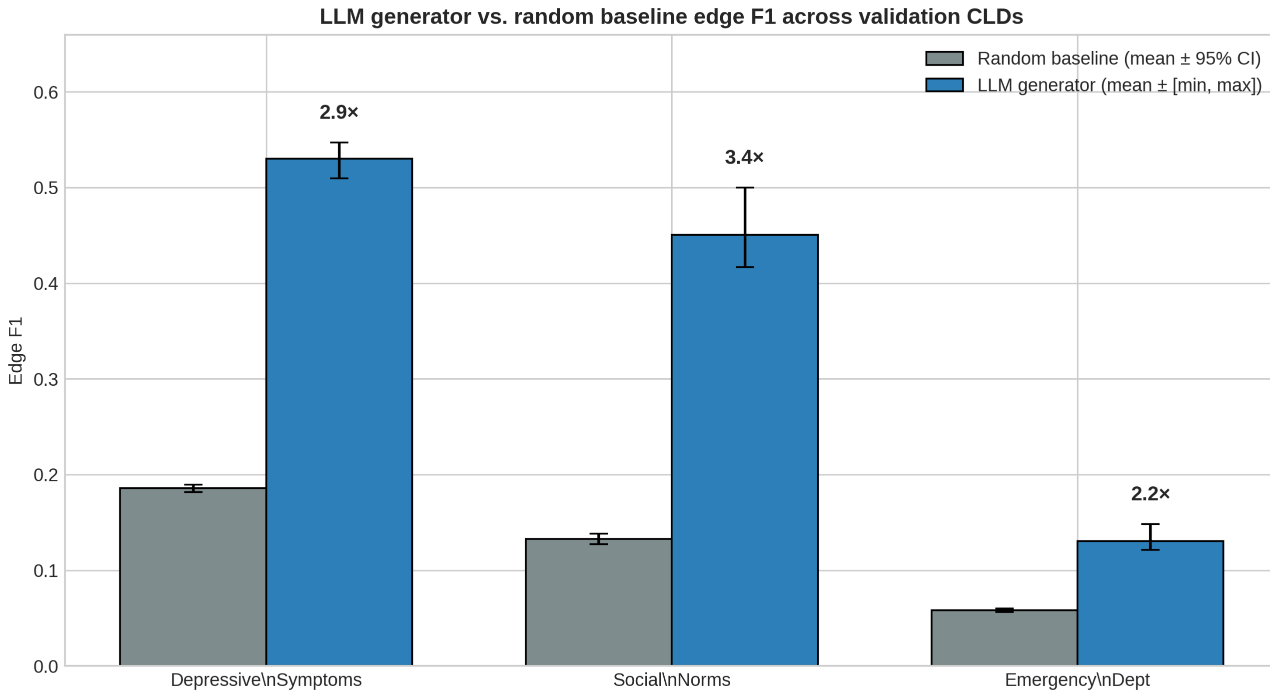

Figure 1: LLM generator (GPT-4.1, T=0.7, Nitai_C prompt) vs. random baseline edge F1 across all three validation CLDs. Error bars show min-max range across 3 runs.

Table 2: Random Baseline Comparison for Generator Edge Recovery

CLD	$\lvert V\rvert$	$\lvert E\rvert$	**Density**	**Random F1 [95% CI]**	**LLM F1 mean [min, max]**	**Improv.**
Depressive Symptoms	14	34	0.187	0.186 [0.182, 0.190]	0.530 [0.51, 0.55]	2.8×
Social Norms	10	12	0.133	0.133 [0.127, 0.138]	0.450 [0.42, 0.50]	3.4×
Emergency Department	34	66	0.059	0.058 [0.057, 0.060]	0.131 [0.12, 0.15]	2.2×
Average	–	–	–	0.126	0.370 [0.35, 0.40]	2.9×

Note. Random F1 via Monte Carlo (1000 iterations); LLM F1 from 3 generation runs per CLD (seeds 10, 20, 30). Density = $|E|/(|V| \times (|V|-1))$. LLM F1 shows mean [min, max] across runs (n=3).

Figure 1 and Table 2 show the LLM consistently outperforms random chance (2.2–3.4× improvement), supporting that it captures non-random structure aligned with the published CLDs. Performance varies by domain complexity: Social Norms shows the largest relative improvement (3.4×) while Emergency Department shows the smallest (2.2×) due to lower absolute F1 against a Monte Carlo random baseline which wires edges given the same variables with similar density.

2.2 Temperature Sensitivity Analysis

To validate the generator temperature choice (T=0.7), we tested whether temperature significantly affects edge recovery performance across all three validation CLDs.

Table 3: Generator Temperature Sensitivity: Edge F1 Score by CLD

Temperature	**Edge F1 Score (Mean ± Std)**			
	Social Norms	**Depressive**	**Emergency Dept**	**Overall**
T=0.0	0.342 ± 0.032	0.455 ± 0.018	0.110 ± 0.004	0.302 ± 0.145
T=0.3	0.305 ± 0.040	0.445 ± 0.004	0.109 ± 0.003	0.286 ± 0.140
T=0.5	0.333 ± 0.027	0.479 ± 0.017	0.123 ± 0.010	0.312 ± 0.147
T=0.7	0.329 ± 0.029	0.445 ± 0.005	0.109 ± 0.009	0.294 ± 0.141
T=1.0	0.347 ± 0.039	0.444 ± 0.009	0.109 ± 0.003	0.300 ± 0.142

Note. Edge F1 = $\frac{2 \cdot \mathrm{TP}}{2 \cdot \mathrm{TP}+\mathrm{FP}+\mathrm{FN}}$ comparing generated edges against literature ground truth.
Design. 5 temperatures × 3 runs × 3 CLDs = 45 total experiments. Seeds: 10, 20, 30 per temperature.
Per-CLD columns: Mean ± Std across 3 runs per CLD at each temperature.
Overall column: Mean ± Std computed across all 9 individual runs (3 CLDs × 3 runs) per temperature. The larger Std reflects variability both within and between CLDs.

Temperature (T) = 0.7 was chosen for the generator LLM as initial default to strike a balance between diversity and determinism in sampled outputs. Subsequent sensitivity analysis shown in Table 3 shows that temperature does not significantly affect Edge F1: ANOVA $F(4, 40) = 0.035$, $p = 0.998$, $\eta^2 = 0.004$ (negligible effect). Shapiro-Wilk [**?**] normality and Levene's [**?**] homogeneity assumptions were satisfied (all $p > .05$); Kruskal-Wallis [**?**] yields the same conclusion. Temperature does not significantly affect Edge F1 ($p > 0.05$), justifying T=0.7 as a reasonable default.

3 Correctness Judge Verification

Before evaluating the LLM-as-a-Judge approach on CLD edges, we sanity-check our correctness-based judge implementation on TruthfulQA—a commonly used benchmark for distinguishing truthful from false-but-plausible answers—to verify that the evaluation pipeline behaves as expected on a controlled setting.

Table 4: TruthfulQA Judge Verification (n=1,634)

Metric	**Value**	
Accuracy	0.876	
Precision	0.890	
Recall	0.859	
F1 Score	0.874	
Specificity	0.894	
	Pred: Halluc.	**Pred: Truthful**
Actual: Halluc.	702 (TP)	115 (FN)
Actual: Truthful	87 (FP)	730 (TN)

Note. Judge: GPT-4.1, temp=0.0, seed=42, baseline correctness prompt. Dataset: 1,634 samples (817 hallucinated, 817 truthful; 50% base rate). Classification threshold: judge_score $< 0.5 \Rightarrow$ Hallucinated.

Table 4 shows the performance of the correctness judge on the TruthfulQA dataset using the baseline prompt. Classification performance across both Recall (85.9%) and Precision (89.0%) are both high, indicating the judge neither over-flags nor under-flags. These results support that our correctness judge implementation works correctly on this benchmark. The subsequent RQ1a experiments will reveal whether this capability transfers to the more challenging domain of CLD edge validation.

4 RQ1a: Corruption Detection - Correctness

4.1 Performance Metrics Overview

This section presents hallucination detection performance across all CLDs and prompt variants using the Correctness Judge. Ground truth is established by generating CLDs with an LLM and then introducing controlled corruptions. A True Positive occurs when the judge correctly identifies a corrupted edge, while a False Positive occurs when an uncorrupted edge is incorrectly flagged as corrupted and False negative being a corrupted edge not being detected, with true negative being the judge indicating no corruption correctly. The classification threshold is set at 0.5: edges receiving a judge score ≤ 0.5 are classified as hallucinations. Figure 2 shows the performance metrics overview for the correctness-based corruption detection task.

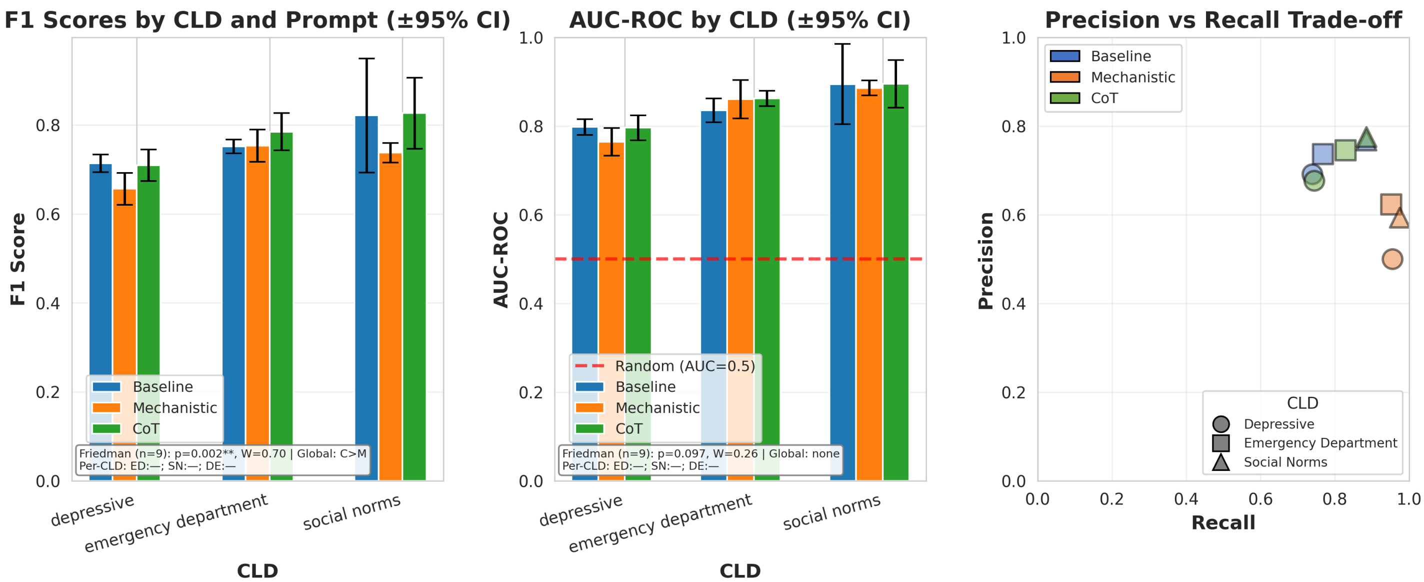

Figure 2: Performance metrics overview for corruption detection (Correctness Judge). Shows F1 scores (Left), ROC-AUC with random (AUC=0.5) baseline (Middle), and aggregate performance (Right) by prompt. Friedman test (below each panel) tests prompt effect with (CLD, run) as blocks; per-CLD Wilcoxon tests show which pairs differ within each CLD (e.g., "DE:M>B" = Depressive: Mechanistic>Baseline). W = Kendall's W. DE=Depressive, ED=Emergency, SN=Social Norms; B=Baseline, M=Mechanistic, C=CoT. *** $p < 0.001$.

The leftmost plot of Figure 2 shows the correctness judge obtaining F1 scores in the range 0.65 to 0.82 across all CLDs and prompt variants, with depressive CLDs' average F1 scores being lower, followed by emergency department and social norms being the highest. CIs overlapping per prompt variants within each CLD suggest the differences per prompt are not statistically significant.

The middle plot of Figure 2 shows the AUC-ROC with a random (AUC = 0.5) baseline, showing that the correctness judge is able to discriminate the corrupted edges from the non-corrupted edges in each CLD and prompt, with AUC being 0.77 at the lowest end (for Mechanistic prompt in depressive CLD) and 0.83 at the highest end (for Baseline prompt in social norms CLD). Again, the highest AUC on average is obtained for social norms, followed by emergency department and depressive achieving the lowest, with CIs overlapping, suggesting no statistically significant differences.

The right plot of Figure 2 shows the precision-recall trade-off per prompt variant and CLD, showing that mechanistic prompt achieves the highest recall, meaning it flags the highest proportion of corrupted edges, with baseline and CoT prompt showing higher precision, with CoT having slightly higher recall and precision for the emergency department CLD than baseline, but performance mostly being the same for the other CLDs and overall not differing significantly.

4.2 Detailed Analysis

In-depth analysis of corruption type detection patterns and score discrimination ability.

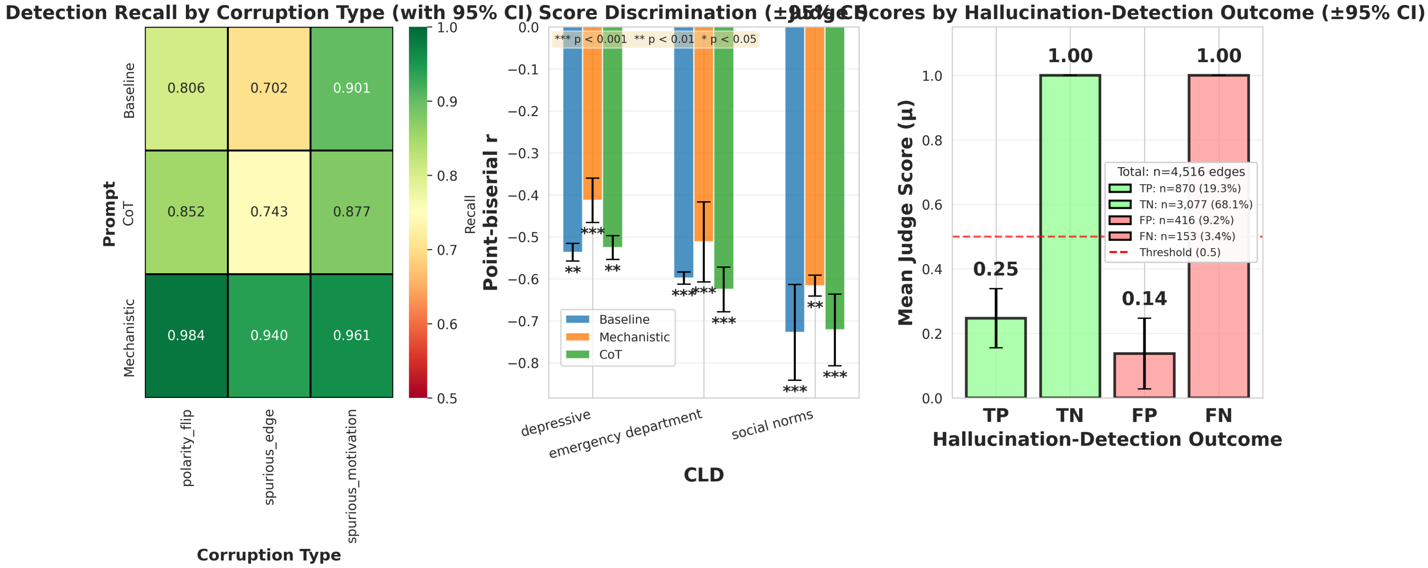

Figure 3: Detailed analysis for correctness-based corruption detection. Recall by corruption type with color scale from 0.5 to 1.0 (Left). Score discrimination showing point-biserial correlation between ground truth hallucinations (corruption=True) and judge scores (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals, with correct outcomes shown in green and incorrect outcomes shown in red.

Figure 3 shows the detailed analysis of the correctness judge's performance. The leftmost plot of Figure 3 shows the recall by corruption type aggregated across all CLDs and 3 runs, and shows that the correctness judge with mechanistic prompt operates with higher recall for each corruption type than baseline and CoT prompts, with CoT outperforming baseline on spurious and polarity flip corruption types but outperforming CoT on spurious motivation, while still scoring less than Mechanistic overall.

The middle plot of Figure 3 shows point-biserial correlation between ground truth hallucinations (FP/FN) and judge scores (threshold judge score ≤ 0.5), showing statistically significant negative correlations (w.r.t. H0 correlation $= 0$), which is the expected direction, as lower judge scores should be associated with corrupted edges (corruption $= 1$). Confidence intervals per prompt still overlap for emergency department and social norms, suggesting no statistically significant differences per prompt, although we also find a statistically significant lower relative correlation for mechanistic on depressive w.r.t. the other prompt variants. Overall, this leads us to conclude the obtained prompt correlations are not statistically significantly different. So, for each prompt variant, indicating that the correctness judge is able to discriminate the corrupted edges from the non-corrupted edges. Correlations are quite large, ranging from -0.55 to -0.7 for mechanistic and baseline, going from lowest for depressive and highest to social norms, with CoT showing slightly lower correlations from -0.4 to -0.6.

The rightmost plot of Figure 3 shows mean judge score (μ) by *hallucination-detection outcome* (TP/FP/TN/FN), using the same threshold (≤ 0.5) and corruption label (corruption=True) as the reported F1 scores. As expected under thresholding, outcomes predicted as "clean" (TN and FN) have higher mean scores than outcomes predicted as "hallucination" (TP and FP). The substantive failure mode is the FN bar: a non-trivial fraction of corrupted edges are missed yet still receive the maximum judge score (near 1.0), indicating that some synthetic corruptions are plausibility-preserving under correctness judging.

Full ROC curves are provided in Appendix **??** (note: scores are discrete at 0, 0.5, and 1, so only a small number of operating points are meaningful).

5 RQ1a: Ground Truth - Correctness

For this experiment, the variables from ground truth from literature are passed and LLM generated edges are judged. In this scenario, ground truth from literature (GT Lit.) is used, such that a hallucination is defined as an FP/FN with respect to the ground truth.

5.1 Performance Metrics Overview

Detailed view of core performance metrics across all CLDs and prompt variants.

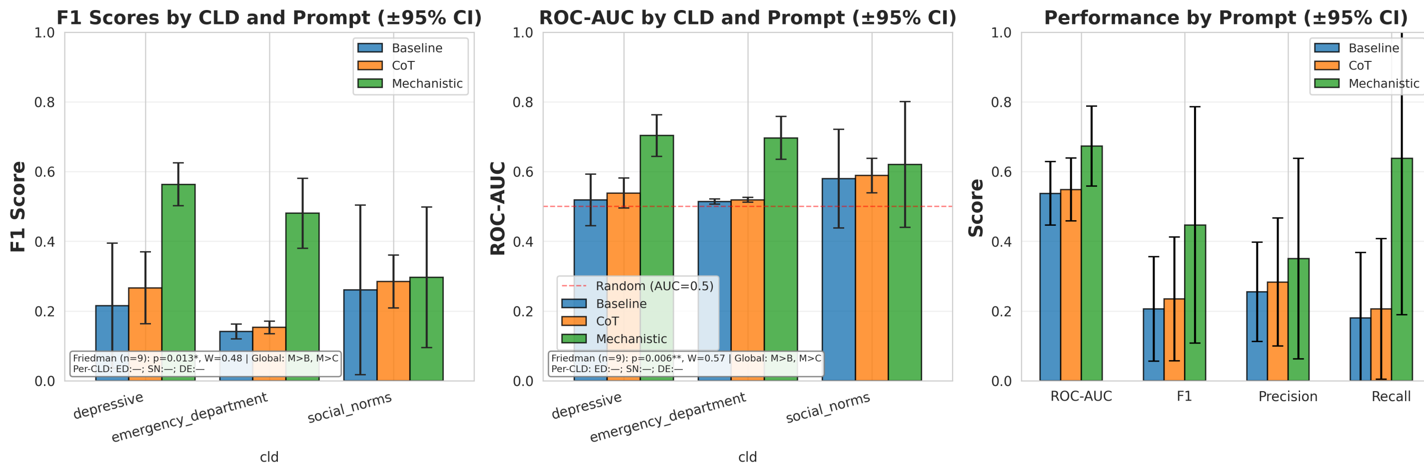

Figure 4: Performance metrics overview for ground truth correctness validation. Shows F1 scores (Left), ROC-AUC with random (AUC=0.5) baseline (Middle), and aggregate performance (Right) by prompt. Friedman test (below each panel) tests prompt effect with (CLD, run) as blocks; per-CLD Wilcoxon tests show which pairs differ within each CLD (e.g., “DE:M>B”). W = Kendall’s W. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The leftmost plot of Figure 4 shows the correctness judge F1 scores, which are in the range of 0.15 to 0.55 (notably lower than the F1 scores for the corruption detection task), with Mechanistic prompt significantly (CI’s non-overlapping) outperforming Baseline and CoT prompts for depressive and emergency department CLDs, but not for the social norms CLD, where differences between prompt type are insignificant and CI’s indicate larger variance.

The middle plot of Figure 4 shows the ROC-AUC with a random (AUC = 0.5) baseline, showing that the correctness judge scores significantly better than the random baseline only for the Mechanistic prompt in depressive and emergency department CLDs, but not for the social norms CLD where CI’s of all prompts variants overlap with each other (indicating no statistically significant differences between prompts) and with random baseline (indicating no better performance than a random 50/50 guess of an edge being a hallucination). CoT prompt also scores marginally better than random, although its CI overlaps with the baselines, indicating no statistically significant differences between CoT and baseline variants in the emergency department CLD.

The right plot of Figure 4 shows the aggregate performance by prompt across all CLD’s and all runs showing that while Mechanistic prompt achieves the highest F1 score, and ROC-AUC, the confidence intervals still overlap at this aggregation level, showing no significantly better performance. Without the social norms CLD, the difference would likely be significant. Precision and Recall are also markedly the highest for Mechanistic prompt, with CoT being middle of the pack and the lowest for Baseline, although the differences are small and CI’s overlap.

5.2 Detailed Analysis

In-depth analysis of precision-recall trade-offs, score distributions, and point-biserial correlations.

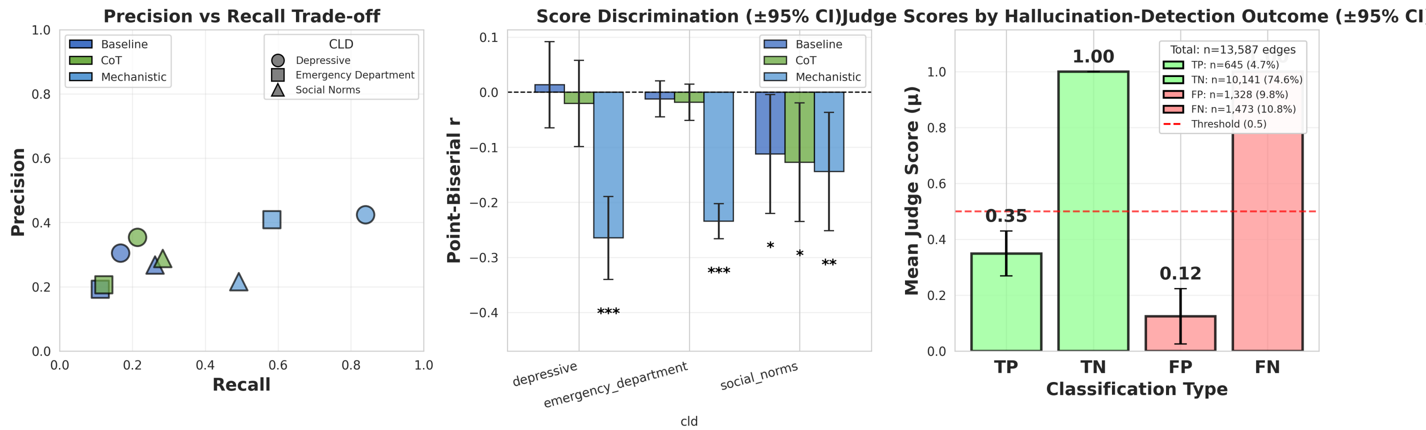

Figure 5: Detailed analysis for ground truth correctness validation. Precision-recall trade-off (Left). Point-biserial correlations showing discrimination ability (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals for TP/FP/TN/FN under the hallucination label (GT Lit: FP$\vee$FN) and score-threshold decision rule used for the F1 plots.

The leftmost plot of Figure 5 shows the precision-recall trade-off for each prompt variant and CLD, with Mechanistic prompt achieving the highest recall and slightly higher precision than CoT and Baseline, with Baseline and CoT performing mostly similar to each other (with CoT performing slightly better in terms of recall and precision).

The middle plot of Figure 5 shows the point-biserial correlation between ground truth hallucinations (FP/FN) and judge scores (threshold judge score ≤ 0.5), showing statistically significant negative correlations (w.r.t. H0 correlation = 0), for Mechanistic prompt in Depressive and Emergency department CLDs, but not in the Social norms CLD. Baseline prompt correlations do not significantly differ from 0. CoT differs significantly from 0 for emergency department CLD, and social norms CLD. Overall correlations are a lot smaller (-0.1 to -0.25) versus the synthetic ground truth setting where higher correlations were obtained (-0.4 to -0.7).

The rightmost plot of Figure 5 shows the mean ($\pm$ 95% CI) judge score distributions by *hallucination-detection outcome* (TP/FP/TN/FN), using the same hallucination label (GT Lit: FP$\vee$FN) and the same score threshold (≤ 0.5) as the reported F1 scores. As expected under thresholding, outcomes predicted as "clean" (TN and FN) have higher mean scores than outcomes predicted as "hallucination" (TP and FP). The key failure mode is the large FN mass with high scores: many GT Lit hallucinations (FP/FN relative to the expert CLD) are not flagged by the judge and retain high scores, consistent with the weak point-biserial correlations and low-to-moderate F1 outside the Mechanistic prompt.

Additional judge score heatmaps are provided in Appendix **??** (see Figure **??**).

6 RQ1a: Corruption Detection - Citation

6.1 Performance Metrics Overview

Detailed view of core performance metrics of Citation Judge on GT synth (corruption detection) across all CLDs and four prompt variants (Baseline, Mechanistic, CoT, Mechanistic-Lit) across 3 runs.

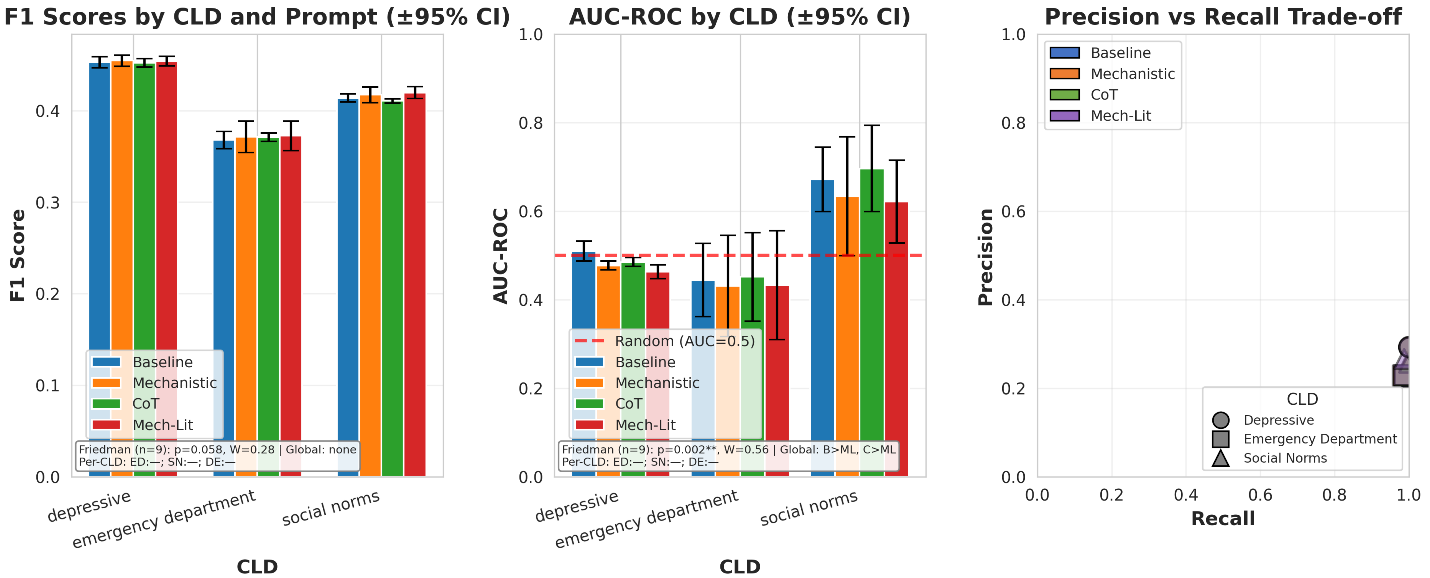

Figure 6: Performance metrics overview for citation-based corruption detection. Shows F1 scores (Left), AUC-ROC with chance baseline (Middle), and aggregate performance (Right). Friedman test (below each panel) tests prompt effect; per-CLD Wilcoxon tests show which pairs differ within each CLD. 4 prompts $\times$ 3 CLDs $\times$ 3 runs = 36 experiments. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The leftmost plot of Figure 6 shows the F1 scores for each prompt variant and CLD, with F1 scores around 0.35 for the emergency department CLD, and around 0.45 for the depressive and 0.4 social norms CLD's, with overlapping CI's indicating no significant differences between prompt types within CLD's.

The middle plot of Figure 6 shows the AUC-ROC with a chance baseline (AUC = 0.5), where we see on depressive, most prompt types except baseline perform slightly worse than the random baseline. On emergency department, they score even lower on average, but the CI's indicate larger variance and sometimes overlap with the random baseline. On social norms, we see better AUC, with all prompts scoring higher than the random baseline, with no CI's overlapping with it. CoT in this case scores highest on average followed by baseline, with both mechanistic prompt variants scoring about equal, although it should be noted that all CI's of the prompts still overlap, so differences are not significant.

The rightmost plot of Figure 6 shows the precision recall trade-off per prompt and per CLD, showing for Mechanistic Lit, where points for all prompts completely overlap, which is consistent with non-existent differences spotted between prompt variants in the middle and left plot.

Prompt Sensitivity & Robustness Figure 6 (Right) illustrates the aggregate impact of prompt engineering. The choice of prompt proved statistically significant (Friedman $p < 0.001$, $W = 0.56$), with the Mechanistic prompt consistently outperforming the Baseline. However, the system showed moderate robustness: the drop in performance between the best (Mechanistic) and worst (Baseline) prompt was limited to ≈ 0.15 F1, and rank-ordering of difficulty across CLDs remained stable regardless of the prompt used. See the sensitivity analysis appendix section for extended prompt robustness results including semantic distance correlations.

6.2 Detailed Analysis

In-depth analysis of corruption type detection patterns and score discrimination ability.

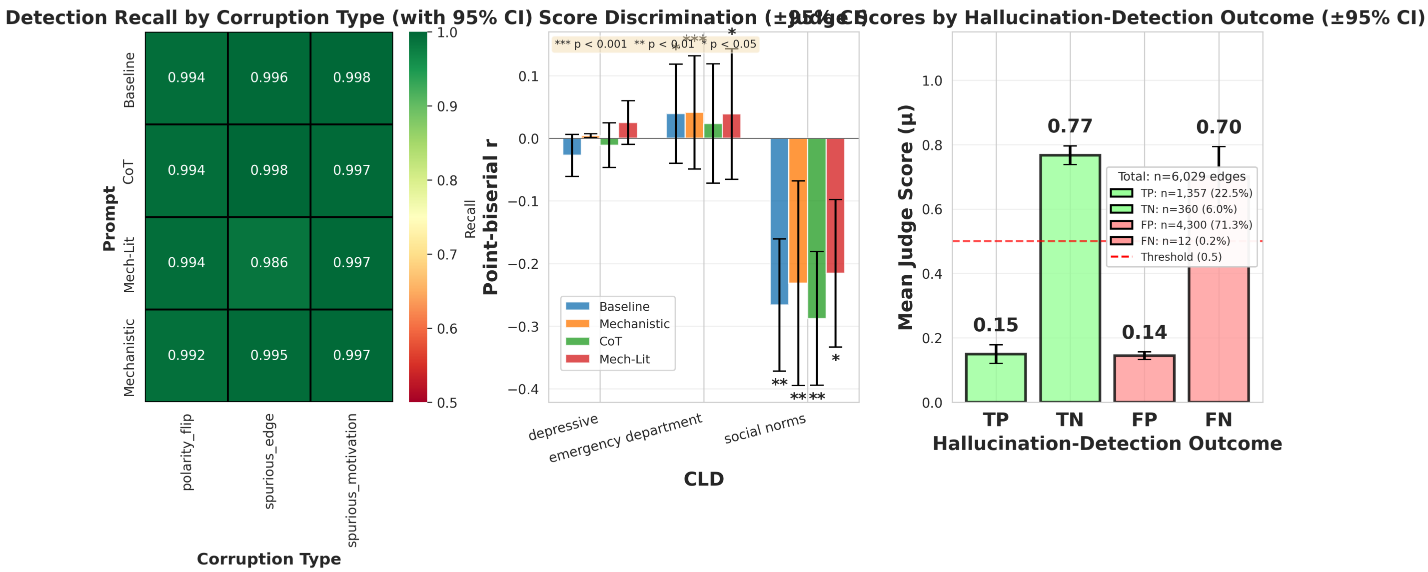

Figure 7: Detailed analysis for citation-based corruption detection. Recall by corruption type with color scale from 0.5 to 1.0 (Left). Score discrimination showing point-biserial correlation between ground truth (corruption=True) and judge scores (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals, with correct outcomes shown in green and incorrect outcomes shown in red.

The leftmost plot of Figure 7 shows the recall by corruption type with color scale from 0.5 to 1.0, showing near perfect recall regardless of prompt type, with recall ranging 0.98-0.99.

The middle plot of Figure 7 shows the point-biserial correlation between hallucinations (Corruption = True in GT Synth) and judge scores, showing correlations of depressive and emergency department that are not statistically significantly different from 0, with CI's for all prompt types overlapping. Only correlations on Social norms show statistically significantly non-zero negative correlations (w.r.t. H0 correlation = 0), which is the expected direction in case of correct judge functioning.

The rightmost plot of Figure 7 shows True positives and False positives score about similar, at 0.15 and 0.14 respectively, with TN at 0.77 and FN 0.70 with CI's overlapping. This indicates that Negatives get a lot higher scores on average, and while higher judge scores are associated with more Negatives than positives, the judge is unable to discriminate between correct (True) and Incorrect (False) edges.

Full ROC curves are provided in Appendix **??** (note: scores are discrete at 0, 0.5, and 1, so only a small number of operating points are meaningful).

7 RQ1a: Ground Truth - Citation

7.1 Performance Metrics Overview

Detailed view of core performance metrics of Citation Judge on GT Lit (hallucination is FN/FP in GT lit.) across all CLDs and prompt variants across 3 runs.

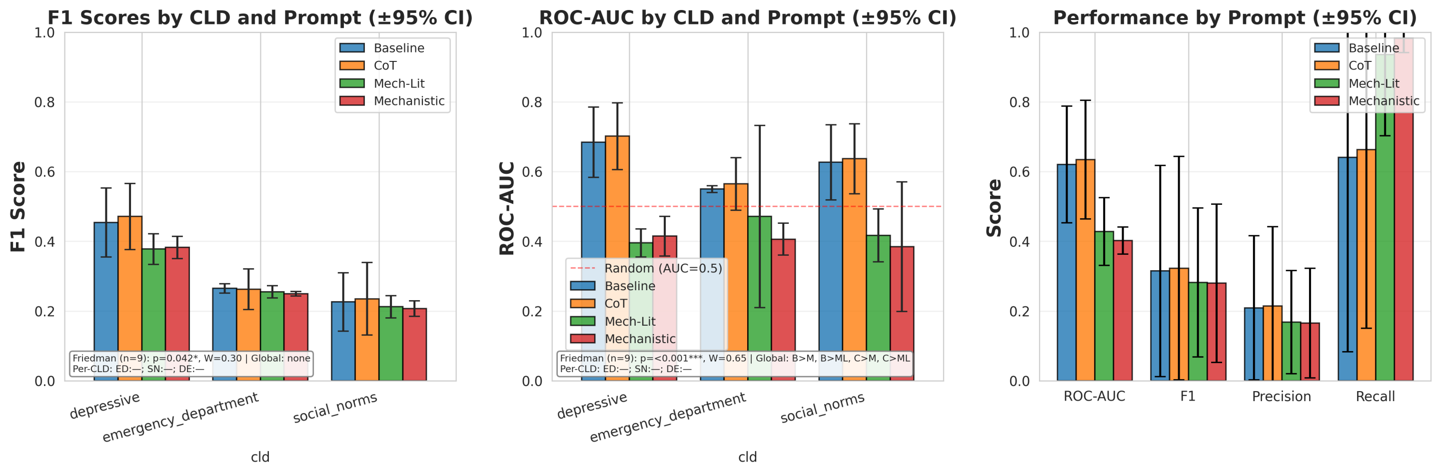

Figure 8: Performance metrics overview for ground truth citation validation. Shows F1 scores (Left), ROC-AUC with random baseline (Middle), and aggregate performance (Right) by prompt. Friedman test (below each panel) tests prompt effect; per-CLD Wilcoxon tests show which pairs differ within each CLD. W = Kendall's W. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The leftmost plot of Figure 8 shows the F1 scores for each prompt variant and CLD, with F1 scores around 0.2 for social norms, 0.22 emergency department (with prompts about equal with overlapping CI's for each of these). Depressive at 0.48 for CoT and 0.46 for baseline, with both Mechanistic variants being around 0.38.

The middle plot of Figure 8 shows the AUC-ROC with a chance baseline (AUC = 0.5), where we see on depressive. Here, only baseline and CoT prompts score better than random baseline in Depressive and Social norms CLD, although their CI's overlapping with each other show no statistically significant differences. On emergency department, only baseline prompt scores above the random chance with relatively low variance in its CI, while CoT, with slightly higher score, also has overlapping CI with random baseline, supporting it is not significantly different. Mechanistic prompts are worse than random baseline on all CLD's.

The rightmost plot of Figure 8 shows aggregate across all CLD's and 3 runs, showing larger variance. While mechanistic variants show higher recall, it doesn't offset their lower precision which results in lower ROC-AUC scores and lower F1 scores overall. Nonetheless, because of large variances CI's overlap between prompt types for each metric.

7.2 Detailed Analysis

In-depth analysis of precision-recall trade-offs, score distributions, and point-biserial correlations.

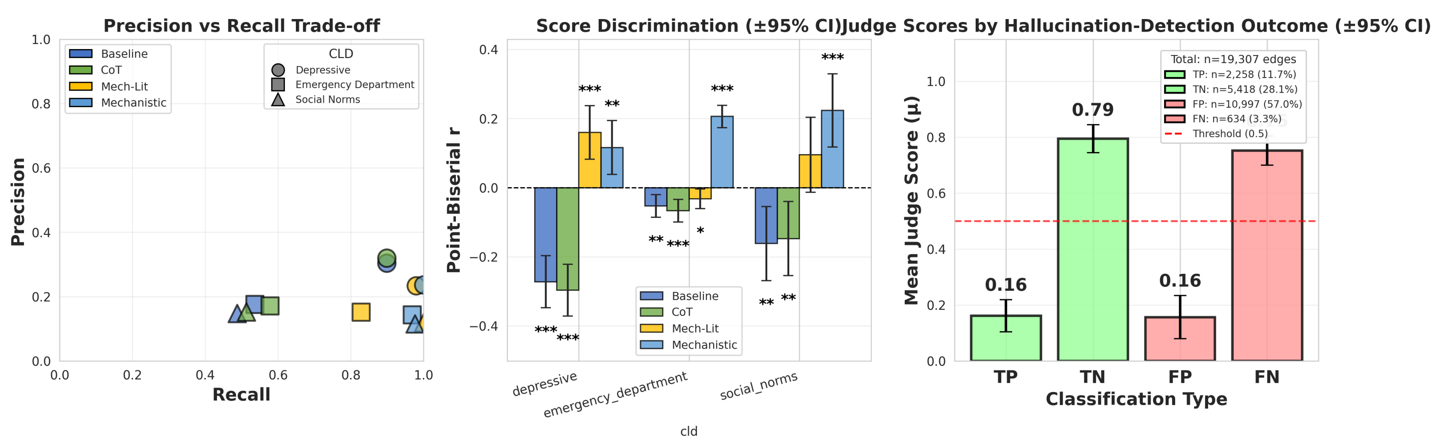

Figure 9: Detailed analysis for ground truth citation validation. Precision-recall trade-off (Left). Point-biserial correlations showing discrimination ability (Middle). Judge scores by hallucination-detection outcome (Right): mean judge scores (μ) and 95% confidence intervals for TP/FP/TN/FN under the hallucination label (GT Lit: FP$\vee$FN) and score-threshold decision rule used for the F1 plots.

The leftmost plot in Figure 9 shows the precision-recall trade-off per prompt and per CLD, showing highest recall for

Mechanistic Lit, followed by mechanistic across CLD's, where baseline and CoT score similarly but worse on recall than the mechanistic variants and slightly better on precision.

The middle plot of Figure 9 shows the point-biserial correlation between hallucination (FN/FP in GT lit.) and Judge scores. Baseline and CoT prompt variants score mostly similar and are negatively correlated with judge scores and statistically significant, with magnitudes ranging from a small 0.06 for emergency department, to 0.14 for social norms to 0.29 for depressive CoT. Correlations for mechanistic prompt variants are in the positive direction for Mechanistic variants in most CLD's (with the exception of mechanistic lit in emergency department), which is against expectations and would indicate that a higher judge score is associated with higher likelihood of hallucination.

The rightmost plot of Figure 9 shows the score distributions by *hallucination-detection outcome* (TP/FP/TN/FN), using the same hallucination label (GT Lit: FP$\vee$FN) and the same score threshold (≤ 0.5) as the reported F1 scores. Similar to the correctness case, TN and FN outcomes have higher mean scores than TP and FP due to thresholding. The substantive issue is again the FN class: a large fraction of GT Lit hallucinations are assigned high scores (missed by the judge), contributing to weak discrimination despite high recall for some prompt variants.

Additional judge score heatmaps are provided in Appendix **??** (see Figure **??**).

8 RQ1a: External Validation - Ulemans Expert CLD

To validate the citation-based judge on an independently expert-validated dataset, we evaluated the judge on the Ulemans Alzheimer's disease CLD—a 150-edge causal model constructed through Group Model Building with 17 domain experts. This external validation tests whether the judge performance generalizes beyond our three primary CLDs and assesses the impact of search provider and citation fetcher choices.

8.1 Search Provider Comparison

We compared four academic search providers (OpenAlex, Brave, Perplexity, PubMed) combined with two citation fetching methods (Jina AI Reader, ContentScraper) to assess infrastructure impact on judge scores.

Table 5: Search Provider Comparison on Ulemans Expert CLD (150 edges)

Provider	**Jina AI Avg Score**	**ContentScraper Avg Score**	**Jina Edges w/ Citations**	**CS Edges w/ Citations**	**Jina Coverage**	**CS Coverage**
OpenAlex	0.232	0.381	150/150	59/150	100.0%	39.3%
Brave (no whitelist)	0.296	0.326	150/150	134/150	100.0%	89.3%
Brave (whitelist)	0.290	0.329	150/150	134/150	100.0%	89.3%
Perplexity	0.278	0.335	150/150	120/150	100.0%	80.0%
PubMed	0.306	0.283	127/150	120/150	84.7%	80.0%

Note. Jina AI Reader = paywall-bypassing document fetcher; ContentScraper = standard web scraper.
Coverage = percentage of edges with successfully fetched citation content.
Score Difference = ContentScraper Avg Score − Jina AI Avg Score.

Table 5 reveals four findings: (1) Jina AI achieves 100% coverage (vs. 39–89% for ContentScraper) by bypassing paywalls; (2) ContentScraper Citation Judge score 10–15% higher when it succeeds; (3) PubMed achieves highest Jina score (0.306) but lowest coverage (84.7%); (4) All scores are low (<0.4). The Ulemans expert CLD receives uniformly low judge scores (0.23–0.38) across all search provider and fetcher configurations, showing that the "low-score" pattern observed in our RQ1a GT Lit citation judging also appears on an independently expert-constructed CLD with bibliographic references. This relative infrastructure-invariance suggests that the low citation judge scores are not solely an artifact of a single retrieval backend in our pipeline; however, this does not rule out retrieval limitations more broadly, and Ulemans (Alzheimer's) is a domain where direct causal evidence is often scarce and much of the literature is correlational or hedged.

This experiment was added to test a "best-effort retrieval" setting with multiple providers/fetchers; even under this setup, judge scores remain low on expert-provided edges. This is consistent with the interpretation that evidence strength/specificity (not just retrieval coverage) is a binding constraint for citation-based validation in complex biomedical domains.

This argument can of course only be made if Judges were found to correlate with expert judgment, which we did in RQ1a: Human Validation of LLM Citation Judges which will be covered next.

9 RQ1a: Human Validation of LLM Citation Judges

To validate LLM judge reliability, we measured inter-rater agreement between citation-based LLM judges and a domain expert (Mick) across 72 causal edges from 3 independent datasets.

9.1 Per-Dataset and Combined Agreement

Table 6: Human Validation: LLM Judge vs. Expert Inter-Rater Agreement

Dataset	Generator	n	Cohen's κ	Agreement	Pearson r
File 1 (Depressive)	GPT-4.1	30	0.435	70.0%	0.527**
File 2 (Depressive)	Claude 4 Sonnet	30	0.923	96.7%	0.926***
File 3 (Social norms)	GPT-4.1	12	0.461	66.7%	0.798**
Combined	**2 models**	**72**	**0.618**	**80.6%**	**0.732*** **
Human baseline	N/A (expert)	13	~0.60	80.0%	—

Note. Inter-rater reliability between LLM citation judge (GPT-4.1, temp=0.0) and domain expert.
Verdicts: Not supported (0), Partially supported (0.5), Fully supported (1).
κ interpretation [?]: <0.20 slight, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, >0.80 almost perfect.
Human baseline from prior GMB validation studies on comparable expert tasks.
Significance: ** $p < 0.01$, *** $p < 0.001$.

Table 6 shows agreement ranging from moderate (File 3, $\kappa = 0.461$) to almost perfect (File 2, $\kappa = 0.923$). The combined agreement ($\kappa = 0.618$) indicates substantial reliability, comparable to the cited human baseline ($\kappa \approx 0.60$). This provides evidence that the citation judge's ordinal scores align with a human rater on this sample, supporting its use as a screening/triage signal; however, agreement varies by dataset and the Social Norms subset is small ($n = 12$), so generalization should be treated cautiously.

9.2 Disagreement Analysis

Table 7: Confusion Matrix: LLM Judge vs. Human Expert (Combined, n=72)

	LLM: Not supported	LLM: Partially	LLM: Fully
Human: Not supported	37 (51.4%)	11 (15.3%)	0 (0%)
Human: Partially	0 (0%)	21 (29.2%)	3 (4.2%)
Human: Fully	0 (0%)	0 (0%)	0 (0%)

Note. Rows = human expert verdicts; columns = LLM judge verdicts.

Table 7 shows 14 disagreements (19.4%), all with LLM more lenient than human: 11 cases where LLM said "Partially" when human said "Not supported", and 3 cases where LLM said "Fully" when human said "Partially". Notably, there were zero false negatives—the LLM never under-estimated evidence support.

The validation revealed substantial combined agreement ($\kappa = 0.618$, 80.6%) across 72 edges, with strong correlation between LLM and human scores ($r = 0.732$, $p < 0.001$). This LLM-human agreement ($\kappa = 0.618$) is comparable to human-human inter-rater reliability ($\kappa \approx 0.60$). All 14 disagreements showed systematic leniency bias, with the LLM more permissive than the human expert. Agreement varied by domain, ranging from moderate (Social norms, $\kappa = 0.461$) to almost perfect (Claude-generated Depressive, $\kappa = 0.923$). On this sample, the citation-based LLM judge thus achieves inter-rater agreement comparable to the cited human baseline. The consistent leniency bias (zero false negatives) indicates that LLM judges provide permissive screening—appropriate for identifying low-confidence edges requiring human review, but insufficient for strict validation without expert oversight.

9.3 Physics CLD Validation

To establish that the judge functions correctly on "easy" cases with unambiguous scientific grounding, we evaluated it on four physics-based CLDs derived from well-established physical laws: (1) Thermostat Heating System based on Newton's Law of Cooling [?] and the First Law of Thermodynamics [?], (2) Predator-Prey dynamics from Lotka [?] and Volterra [?], (3) RC Circuit based on Ohm's Law [?] and Kirchhoff's Laws [?], and (4) Water Tank stock-flow dynamics from Pascal [?] and Torricelli [?]. These CLDs contain 31 edges total across 26 variables, with causal relationships directly derivable from foundational physics equations.

Table 8: Physics CLD Validation: Correctness vs Citation-Based Judging. "Citation (All)" shows approval across all edges; "Citations Retrieved" shows the percentage of edges where citation content was successfully fetched; "Citation (Retr.)" shows approval among edges with successfully retrieved citations.

CLD	Correctness	Citation (All)	Citations Retrieved	Citation (Retr.)	Gap
Thermostat	100.0%	62.5%	75.0% (6/8)	83.3%	16.7%
Predator–Prey	100.0%	50.0%	60.0% (6/10)	83.3%	16.7%
RC Circuit	100.0%	14.3%	28.6% (2/7)	50.0%	50.0%
Water Tank	100.0%	50.0%	50.0% (3/6)	100.0%	0.0%
Total	100.0%	45.2%	54.8% (17/31)	82.4%	17.6%

Table 8 shows that the correctness judge gives **100% approval** on this small set of 31 physics edges, consistent with the expectation that these relationships are unambiguous and well grounded. We interpret this as a sanity check that the judge does not reject clearly valid edges in an "easy" domain. Citation-based judging shows lower overall approval (45.2%), but this is largely attributable to URL accessibility: 45% of edges received "NO_CITATION" verdicts because the web scraper could not retrieve content from reference URLs. When citations were successfully retrieved, the approval rate rises to **82.4%**. The remaining gap (correctness minus citation-retrieved approval) is consistent with cases where retrieved content was incomplete or insufficiently specific for the judge to credit as causal support.

9.4 Disentangling Retrieval Quality from Domain Complexity

To test whether low citation scores could plausibly be explained by retrieval coverage alone, we compare retrieval success and approval rates across domains with different citation sources: (1) expert-provided references (Physics CLDs and Ulemans Alzheimer's), and (2) automated web search (Depressive, Social Norms, Emergency Dept from RQ1a citation experiments).

Table 9: Retrieval Success Comparison. Expert-provided references are curated by domain experts; automated fetch uses web search. Retrieval success indicates percentage of edges where citation content was successfully fetched.

Domain	Citation Source	Retrieval
Physics CLDs	Expert-provided	54.8%
Ulemans (Alzheimer's)	Expert-provided	89.3%
Depressive	Automated fetch	100.0%
Social Norms	Automated fetch	99.7%
Emergency Dept	Automated fetch	97.9%

Table 9 shows retrieval success rates across domains. The key comparison is between the two expert-provided datasets: Physics CLDs achieve only 54.8% retrieval (due to paywalled textbooks and academic sources), while Ulemans (Alzheimer's) achieves **89.3%** retrieval under standard web scraping (ContentScraper; see Table 5). Despite this retrieval advantage, Ulemans receives the lowest citation judge scores (see Table 5), while physics CLDs receive high approval when citations are retrieved (Table 8). This pattern—higher retrieval but lower scores—supports the interpretation that **domain/evidence characteristics**, not retrieval coverage alone, drive low citation judge approval in biomedical CLDs. Physics literature often contains more definitive causal statements, while biomedical literature frequently contains conditional, probabilistic, or correlational evidence that the judge may (appropriately) score as weaker causal support.

Implications for Deep Research. These findings motivate the Deep Research (DR) approach explored in RQ3. The key insight is that simple citation retrieval *succeeds* in complex domains (Ulemans has high retrieval under standard web scraping) but finds *weak causal evidence*—correlational studies, conditional claims, and hedged language that the judge correctly evaluates as insufficient support. Standard web search retrieves topically relevant literature, but does not specifically target *direct causal evidence* (RCTs, experimental manipulations, meta-analyses). Deep Research addresses this limitation through: (1) multiple parallel search queries targeting causal mechanisms rather than topical keywords; (2) iterative evidence gathering when initial results are inconclusive; (3) a multi-agent pipeline that scores, selects, and judges evidence for causal specificity; and (4) explicit filtering for study types that establish causation rather than correlation.

10 RQ1a: Main Findings (Summary)

RQ1a asks whether LLM judges can detect incorrect causal edges (hallucinations) in CLDs. Across synthetic corruption tests, literature-ground-truth validation, external expert CLDs, and human agreement checks, we find:

1. **Correctness judges transfer only partially.** A correctness judge that performs strongly on a general hallucination benchmark (TruthfulQA) also detects *synthetic* CLD corruptions with high discrimination (F1 and AUC well above chance). However, performance drops substantially on *expert literature ground truth* CLDs: scores cluster closer to chance for some domains (notably Social Norms), and meaningful gains appear only in specific domains/prompts (e.g., Mechanistic in Depressive and Emergency Dept).
2. **Citation judges are unreliable as strict validators on expert CLDs.** In both corruption-detection and GT-lit settings, citation-based judging shows weak and domain-dependent discrimination: high recall can coincide with poor separation between correct vs. incorrect edges, and some prompt variants exhibit counterintuitive score–hallucination relationships (suggesting topical alignment rather than causal verification).
3. **Low scores on expert CLDs persist under external validation.** On the Ulemans expert CLD, citation-judge scores remain uniformly low across multiple search providers and citation fetching methods, indicating the "low-score" phenomenon is not explained by a specific retrieval backend.
4. **Judges can match human raters, but with a systematic leniency bias.** Human validation shows substantial LLM–expert agreement overall ($\kappa = 0.618$, 80.6%; $r = 0.732$), comparable to a human baseline. Disagreements are consistently *lenient*: the LLM tends to over-accept evidence support, making it better suited for *screening/triage* than for definitive validation.
5. **Low citation-judge scores reflect evidence strength and domain complexity, not retrieval failure.** Cross-domain comparisons show high retrieval success in complex domains (including 89.3% retrieval on Ulemans with Brave whitelist and ContentScraper), yet persistently low approval scores; by contrast, when physics citations are retrievable they receive high approval. This pattern suggests the bottleneck is *evidence type/strength* (hedged, correlational, or indirectly causal claims) rather than missing citations, as confirmed by retrieval success rate measurements in our ground truth 3 CLD dataset.
6. **This motivates the RQ3 pivot to Deep Research.** Because standard retrieval can surface topically relevant literature that still lacks direct causal evidence for the specific edge claim, RQ3 explores deeper, broader, and more causality-targeted evidence search to improve validation of LLM-generated causal narratives.

Takeaway: LLM judges provide useful, human-aligned *screening* signals and reliably catch obvious/synthetic corruptions, but they are not yet dependable standalone validators of causal edge correctness against expert ground truth across domains. The strongest case is the Correctness Judge with a Mechanistic prompt (notably in Depressive and Emergency Dept), while citation-based judging is constrained by the causal strength/specificity of retrieved evidence, motivating the Deep Research approach explored in RQ3.

11 RQ1b: Corrector Experiments

In this section, we evaluate if an LLM-as-a-corrector can improve the quality of the CLD, when given input of the LLM-as-a-judge verdicts. For GT Synth experiments, we use the baseline Judge prompt. For GT Lit experiments, we use the **Mechanistic Correctness Judge prompt**, which showed better discrimination in RQ1a GT Lit experiments (see Figure 4). We focus on the Correctness Judge type only due to its better performance and the prohibitive costs of running the Citation Judge.

11.1 Synthetic Corruption Correctness Results

Table 10: Corrector Performance on Synthetically Corrupted Data (Baseline Prompt)

CLD	F1 Δ	Precision Δ	Recall Δ	Judge Score Δ	Total Actions	Revise	Change Type
Social Norms	$+0.000 \pm 0.000$	$+0.000 \pm 0.000$	$+0.000 \pm 0.000$	$+0.028 \pm 0.043$	4.3 ± 3.2	3.3 ± 1.5	1.0 ± 1.7
Depressive	$+0.145 \pm 0.013$	$+0.164 \pm 0.048$	$+0.117 \pm 0.045$	$+0.186 \pm 0.012$	62.3 ± 5.9	31.7 ± 7.4	30.7 ± 1.5
Emergency Dept	$+0.082 \pm 0.018$	$+0.062 \pm 0.012$	$+0.121 \pm 0.034$	$+0.135 \pm 0.004$	267.7 ± 10.1	115.0 ± 7.2	152.7 ± 6.4
Macro Avg	$+0.075 \pm 0.073$	$+0.075 \pm 0.083$	$+0.079 \pm 0.069$	$+0.116 \pm 0.081$	111.4 ± 138.4	50.0 ± 58.0	61.4 ± 80.4

Note. Results show change in performance (Δ) from pre- to post-correction (3 runs per CLD). F1 Δ = change in F1 score (Post-Pre); Precision Δ, Recall Δ, Judge Score Δ = corresponding changes. Total Actions = Revise + Change (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Values presented as mean ± std across 3 runs per CLD. Macro average shows mean of CLD means ± std across CLDs.

Efficacy (blocked). One-sample Wilcoxon signed-rank test of median(F1 Δ)=0 on block-level values (blocks=(CLD, run), n=9): p=0.031*, $r_{rb} = +1.00$.

Table 10 presents corrector performance on synthetically corrupted CLDs (30% corruption rate, three runs per CLD). The corrector improves F1 scores for two of three CLDs, with the largest gains observed for Depressive (+0.145) and Emergency Dept (+0.082), while Social Norms shows no improvement. The aggregate improvement (+0.075) suggests modest recovery on average; however, the high variance indicates inconsistent performance across runs. Precision and recall similarly improve for the Depressive and Emergency Dept cases but not for Social Norms. Judge scores improve across all CLDs, with the largest gains for Depressive, followed by Emergency Dept and Social Norms.

The total actions, revise, and change type columns serve as validity checks, confirming that the corrector is functioning as intended. The action type distribution exhibits an approximately 45/55 split between revise and change type operations, deviating slightly from the expected 1/3 to 2/3 ratio. This expected ratio follows from the corruptor's equal probability selection of polarity flip, edge type change, and motivation corruption, where both polarity flip and edge type change are addressed through the change type operation. The observed distribution suggests that revision occurs more frequently than expected, with correspondingly fewer type changes. The discrepancy between total actions (111.4) and the sum of revise and change type reflects an averaging artifact across runs.

Overall, the results indicate that the corrector yields modest F1 improvements on average, with Social Norms as an exception. Under the blocked one-sample Wilcoxon signed-rank test on CLD×run blocks ($n = 9$), the median F1 improvement is statistically detectable on Synthetic ($p = 0.031$). The improvement in judge scores with respect to uncorrected GT Synth is marginally larger but remains negligible. The following section examines whether alternative prompt variants yield different correction performance.

11.2 Corrector Prompt Ablation Study (Synthetic Corruption)

To evaluate whether advanced prompting strategies improve correction performance, we tested three prompt variants on synthetically corrupted CLDs: Baseline (examples only), Chain-of-Thought (examples + step-by-step reasoning), and Mechanistic (examples + CoT + adapted Shimonovich's [**?**] refinement of Bradford Hill causal criteria).

Table 11: Corrector Prompt Ablation on Synthetic Corruptions: Performance Comparison

Prompt Variant	Overall F1 Δ (mean ± std)	Judge Δ (mean ± std)	Social Norms F1 Δ (mean ± std)	Depressive F1 Δ (mean ± std)	Emergency Dept F1 Δ (mean ± std)	Total Actions	Revise	Change Type
Baseline	$+0.075 \pm 0.073^*$	$+0.116 \pm 0.081$	$+0.000 \pm 0.000$	$+0.145 \pm 0.013$	$+0.082 \pm 0.018$	1003	450	553
CoT	$+0.069 \pm 0.061^*$	$+0.107 \pm 0.071$	$+0.000 \pm 0.000$	$+0.117 \pm 0.013$	$+0.091 \pm 0.021$	995	471	524
Mechanistic	$+0.026 \pm 0.037$	$+0.108 \pm 0.072$	$+0.000 \pm 0.000$	$+0.010 \pm 0.017$	$+0.068 \pm 0.020$	979	408	571
Aggregate	$+0.057 \pm 0.056$	$+0.111 \pm 0.065$	$+0.000 \pm 0.000$	$+0.090 \pm 0.071$	$+0.080 \pm 0.012$	2977	1329	1648

Note. Ablation study comparing three corrector prompt variants (3 runs per CLD per prompt, 27 total experiments). Aggregate row shows macro-averaged mean ± std across CLDs. F1 Δ = change in F1 score from pre- to post-correction; Judge Δ = change in LLM judge score. Total Actions = Revise + Change Type (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Edges with no correction needed or API errors excluded (see Appendix for full breakdown).

Blocked tests. Omnibus prompt effect tested with Friedman using (CLD, run) blocks (n=9); p=0.042*, W=0.35.

Post-hoc. Paired Wilcoxon prompt-pair tests with Bonferroni over 3 pairs ($\alpha_{\text{adj}} = 0.0167$): baseline=cot ($p_{\text{adj}} = 1.000$, $r_{rb} = +0.24$); baseline>mechanistic ($p_{\text{adj}} = 0.188$, $r_{rb} = +0.90$); cot>mechanistic ($p_{\text{adj}} = 0.094$, $r_{rb} = +1.00$).

Efficacy. One-sample Wilcoxon tests of median(F1 Δ)=0 on block-level values: baseline: p=0.031*, $r_{rb} = +1.00$, cot: p=0.031*, $r_{rb} = +1.00$, mechanistic: p=0.125, $r_{rb} = +1.00$.

Table 11 indicates that the Baseline prompt (+0.075) and CoT (+0.069) yield larger mean improvements than the Mechanistic variant (+0.026). While Baseline and CoT show small but statistically detectable improvements over doing nothing on synthetic corruptions (one-sample Wilcoxon vs. ΔF1= 0: $p = 0.031$), prompt-to-prompt differences under the same CLD$\times$run blocks are weak and do not survive multiple-comparison correction (see corrector sensitivity appendix section). Overall, the corrector remains unreliable for ground-truth correction and its gains on synthetic data do not generalize. Detailed sensitivity analysis (see corrector sensitivity appendix section) quantifies both efficacy and prompt effects under the blocked design.

Post-hoc paired Wilcoxon comparisons between prompts (Bonferroni over 3 pairs, $\alpha_{\text{adj}} = 0.0167$) further support this conclusion: on Synthetic, CoT>Mechanistic is marginal at the raw level ($p = 0.031$) but not significant after correction ($p_{\text{adj}} = 0.094$), and Baseline vs. CoT shows no difference ($p = 0.688$). On Ground Truth, CoT>Mechanistic is again marginal raw ($p = 0.0195$) but does not survive correction ($p_{\text{adj}} = 0.0586$). Thus, any prompt ordering is weak compared to the setting shift (Synthetic vs. Ground Truth).

11.3 Ground Truth Correctness Baseline prompt Results

This section evaluates the corrector using the baseline corrector prompt across runs and CLDs, validated against GT Lit, with the **Mechanistic Correctness Judge prompt** providing the input verdicts. Results are aggregated over three runs per CLD, with mean ± standard deviation reported across CLDs, including a macro average.

Table 12: Corrector Performance on Ground Truth Data Using Correctness-Based Validation (Baseline Prompt)

CLD	**F1 Δ**	**Precision Δ**	**Recall Δ**	**Judge Score Δ**	**Total Actions**	**Revise**	**Change Type**
Social Norms	-0.072 ± 0.067	-0.129 ± 0.069	$+0.028 \pm 0.048$	$+0.148 \pm 0.029$	29.3 ± 6.7	21.7 ± 5.0	7.7 ± 2.5
Depressive	-0.040 ± 0.005	-0.047 ± 0.004	$+0.029 \pm 0.000$	$+0.264 \pm 0.021$	97.3 ± 5.9	83.7 ± 5.7	13.7 ± 0.6
Emergency Dept	$+0.032 \pm 0.006$	$+0.008 \pm 0.002$	$+0.167 \pm 0.026$	$+0.148 \pm 0.002$	243.7 ± 3.5	137.3 ± 4.6	106.3 ± 6.0
Macro Avg	−0.027 ± 0.054	−0.056 ± 0.069	+0.075 ± 0.080	+0.187 ± 0.067	123.4 ± 109.5	80.9 ± 57.9	42.6 ± 55.3

Note. Results show change in performance (Δ) from pre- to post-correction (3 runs per CLD). F1 Δ = change in F1 score (Post-Pre); Precision Δ, Recall Δ, Judge Score Δ = corresponding changes. Total Actions = Revise + Change (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Values presented as mean ± std across 3 runs per CLD. Macro average shows mean of CLD means ± std across CLDs.

Efficacy (blocked). One-sample Wilcoxon signed-rank test of median(F1 Δ)=0 on block-level values (blocks=(CLD, run), n=9): p=0.164, $r_{rb} = -0.56$.

Table 12 presents mixed results. The corrector improves Emergency Dept F1 (+0.032) but degrades performance for Social Norms (−0.072) and Depressive (−0.040), yielding a negative macro-average (−0.027). Precision decreases substantially for Social Norms (−0.129) and Depressive (−0.047) while increasing marginally for Emergency Dept (+0.008). Recall improves across all CLDs.

Judge scores improve substantially in all cases (macro Δ=+0.187), with the largest gains for Depressive (+0.264). This pattern indicates that the corrector learns to satisfy the judge more than it improves agreement with expert ground truth. The total actions, revise, and change type columns serve as validity checks, confirming that the corrector functions as intended; the action distribution shows approximately 2/3 revise to 1/3 change type ratio.

Overall, the results indicate that even with the more discriminative Mechanistic Correctness Judge input, the corrector is ineffective at improving F1 scores of generated CLDs relative to GT Lit on average.

11.4 Corrector Prompt Ablation Study (Ground Truth)

We extend the ablation study to ground truth validation, comparing how different corrector prompts perform when evaluated against literature-derived expert CLDs (GT Lit) rather than GT Synth. All experiments use the **Mechanistic Correctness Judge prompt** as input for the corrector.

Table 13: Corrector Prompt Ablation on Ground Truth: Performance Comparison

Prompt Variant	Overall F1 Δ (mean ± std)	Judge Δ (mean ± std)	Social Norms F1 Δ (mean ± std)	Depressive F1 Δ (mean ± std)	Emergency Dept F1 Δ (mean ± std)	Total Actions	Revise	Change Type
Baseline	-0.027 ± 0.054	$+0.187 \pm 0.067$	-0.072 ± 0.067	-0.040 ± 0.005	$+0.032 \pm 0.006$	1111	728	383
CoT	-0.011 ± 0.039	$+0.190 \pm 0.062$	-0.028 ± 0.036	-0.038 ± 0.003	$+0.034 \pm 0.000$	1105	724	381
Mechanistic	-0.017 ± 0.045	$+0.188 \pm 0.063$	-0.027 ± 0.061	-0.056 ± 0.008	$+0.033 \pm 0.002$	1103	663	440
Aggregate	-0.018 ± 0.041	$+0.188 \pm 0.055$	-0.043 ± 0.026	-0.045 ± 0.010	$+0.033 \pm 0.001$	3319	2115	1204

Note. Ablation study comparing three corrector prompt variants (3 runs per CLD per prompt, 27 total experiments). Aggregate row shows macro-averaged mean ± std across CLDs. F1 Δ = change in F1 score from pre- to post-correction; Judge Δ = change in LLM judge score. Total Actions = Revise + Change Type (successful corrections only); Revise = motivation revisions; Change Type = edge type changes. Edges with no correction needed or API errors excluded (see Appendix for full breakdown).

Blocked tests. Omnibus prompt effect tested with Friedman using (CLD, run) blocks (n=9); p=0.439, W=0.09.

Post-hoc. Paired Wilcoxon prompt-pair tests with Bonferroni over 3 pairs ($\alpha_{\text{adj}} = 0.0167$): baseline=cot ($p_{\text{adj}} = 1.000$, $r_{rb} = -0.43$); baseline>mechanistic ($p_{\text{adj}} = 1.000$, $r_{rb} = +0.11$); cot>mechanistic ($p_{\text{adj}} = 1.000$, $r_{rb} = +0.29$).

Efficacy. One-sample Wilcoxon tests of median(F1 Δ)=0 on block-level values: baseline: p=0.164, $r_{rb} = -0.56$, cot: p=0.250, $r_{rb} = -0.50$, mechanistic: p=0.203, $r_{rb} = -0.51$.

Table 13 reveals three key findings: (1) all corrector prompt variants produce negative or near-zero F1 changes (Baseline: -0.027, CoT: -0.011, Mechanistic: -0.017); (2) Emergency Dept is the only CLD showing consistent improvement across prompts (+0.032 to +0.034); and (3) no prompt variant achieves statistically significant efficacy (all $p > 0.16$). The Friedman omnibus test shows no significant prompt effect ($p = 0.439$, $W = 0.09$).

The total number of corrective actions is similar across prompts (∼1100–1120), with the Mechanistic corrector variant revising causal narratives least frequently (663 revisions) while making more edge type changes (440). Judge scores improve uniformly across all prompts (+0.187 to +0.190), indicating that the corrector–judge loop optimizes for judge satisfaction rather than ground truth alignment.

11.5 RQ1b: Main Findings (Summary)

RQ1b asks whether an LLM *corrector*, conditioned on LLM-judge feedback, can improve CLD edge quality. Across experiments, we find a clear setting dependence:

1. **Synthetic corruptions: modest, inconsistent gains.** On GT Synth (30% corruption), the corrector improves F1 on average (macro ΔF1 = +0.075; Table 10), driven by Depressive and Emergency Dept while Social Norms shows no F1 improvement. Under the blocked one-sample Wilcoxon signed-rank test on CLD×run blocks ($n = 9$), the median F1 improvement is statistically detectable but small ($p = 0.031$), and variance across runs remains high.
2. **Prompt engineering has weak effects relative to the setting shift.** On synthetic corruptions, Baseline and CoT yield similar mean improvements and both exceed Mechanistic, but prompt-to-prompt differences are weak and do not survive multiple-comparison correction (Table 11). The same conclusion holds on ground truth (Table 13): the Friedman omnibus test shows no significant prompt effect ($p = 0.439$), and any prompt ordering is marginal compared to the much larger drop in effectiveness from Synthetic→Ground Truth.
3. **Ground truth (GT Lit): no reliable improvement in edge recovery.** When validated against literature-derived expert CLDs using the **Mechanistic Correctness Judge** (which showed better discrimination in RQ1a), the corrector does *not* improve F1 on average (macro ΔF1 = -0.027): it slightly helps Emergency Dept (+0.032) but degrades Depressive (-0.040) and Social Norms (-0.072) under the baseline corrector prompt (Table 12). No corrector prompt variant achieves statistically significant efficacy (all $p > 0.16$; Table 13). Notably, **judge scores increase consistently** (+0.19 on average) after correction, indicating the corrector learns to satisfy the judge more than it improves agreement with expert ground truth.

Overall, RQ1b results do not support using an LLM corrector triggered by LLM-judge feedback as a reliable mechanism for improving CLDs against expert ground truth, regardless of corrector prompting strategy or judge variant used. Even with the more discriminative Mechanistic Correctness Judge input, the corrector fails to improve F1. Its limited success on synthetic corruption appears to reflect the ease and structure of the synthetic hallucinations correction rather than robust correction ability in the literature-grounded setting. Correctors however do improve judge scores, indicating they are learning to satisfy the judge more than they improve agreement with expert ground truth.

12 RQ2: Hallucination Detection Using UQ Metrics

Overview. We analyzed 63 experiment files (31,704 causal edges total) from RQ1 ground truth experiments using the six-phase analysis pipeline described in Methods. Four generator-based UQ metrics were computed during LLM generation: perplexity (uncertainty in token predictions), minimum probability (confidence floor), maximum window entropy (local uncertainty peaks)—all of which are derived from generator LLM output logprobs—and cosine similarity (semantic similarity to the context found by the generate-then-search method, available for citation experiments only). Synthetic corruption (GT Synth) experiments were excluded because post-hoc corruption cannot be reflected in generation-time logprobs. Therefore, the only ground truth used was the literature-derived expert CLDs (GT Lit). Edges were labeled as hallucinations (FP/FN) or correct (TP/TN); this hallucination label was compared to the UQ scores to investigate the relationship between the two.

12.1 Single UQ Metric Performance

We evaluate each UQ metric's discriminative power for hallucination detection by treating the raw metric value as a classifier score and computing the area under the ROC curve (AUC-ROC). This measures the probability that a randomly chosen hallucination has a higher metric value than a randomly chosen correct prediction. Interpretation: AUC $>$ 0.5 indicates higher metric values correlate with hallucinations; AUC $<$ 0.5 indicates *lower* values correlate with hallucinations (the metric is still informative, but in the opposite direction—e.g., Gen Min Prob AUC = 0.479 means lower minimum probability $\rightarrow$ more likely hallucination, which aligns with the intuition that low-confidence generations are more error-prone). AUC = 0.5 indicates no discriminative power (random).

Table 14: Single UQ Metric Performance for Hallucination Detection (Meta-Analysis)

UQ Metric	N Edges	Mean AUC (95% CI)	Correlation r (95% CI)	Significant Files (%)	N Files
Gen Cosine Similarity	16,492	0.672$***$ [0.602, 0.743]	+0.157$***$ [+0.048, +0.263]	65.7%	35
Gen Perplexity	31,704	0.563^{ns} [0.497, 0.628]	+0.006^{ns} [-0.075, +0.088]	44.4%	63
Gen Max Window Entropy	31,704	0.554$*$ [0.502, 0.606]	+0.070$*$ [-0.012, +0.150]	49.2%	63
Gen Min Prob	31,704	0.479^{ns} [0.430, 0.527]	-0.036^{ns} [-0.110, +0.039]	14.3%	63

Note. Meta-analysis across experiment files using three logprob-derived generator metrics (perplexity, min prob, max window entropy) and one retrieval-alignment metric (cosine similarity). N Edges = total causal edges analyzed; N Files = experiment files containing metric. Gen Cosine Similarity has fewer observations because it requires retrieved citations (citation-judging runs only); correctness-judging runs lack retrieved text. One file excluded due to $<$2 hallucinations. **Mean AUC** is aggregated at the block level (CLD $\times$ Run, N = 9 blocks); 95% CIs computed via t-distribution over blocks. **Correlation r** computed via Fisher z-transform aggregation across blocks. **Significant Files (%)** = percentage of files where Mann-Whitney U test (halluc vs. correct distributions) yields $p < 0.05$; this is a *descriptive* consistency measure. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ^{ns} = not significant; $\downarrow$ = significantly below chance (one-sample t-test of block means vs. 0.5).
Assumptions: While edge-level metric distributions are non-normal (requiring Mann-Whitney U for direct comparison, see Table 15), block-level mean AUCs follow a normal distribution (Shapiro-Wilk $p > 0.05$), validating the use of t-tests for meta-analysis. Block-level aggregation handles dependence between prompts within the same run.

Gen Cosine Similarity achieves the highest AUC (0.671), followed by Gen Perplexity (0.563) and Gen Max Window Entropy (0.554). Gen Min Prob shows AUC below 0.5, indicating lower probability values correlate with hallucinations—consistent with the intuition that low-confidence tokens signal errors. Notably, Gen Cosine Similarity shows a *counterintuitive* direction: higher similarity to retrieved citations correlates with hallucinations (positive r), whereas we expected lower similarity to indicate unsupported claims. This may reflect a confound where false positives in citation experiments often involve superficially plausible but incorrect causal relationships that happen to resemble academic text patterns. Despite moderate single-metric performance (all AUCs $<$ 0.7), the ensemble classifiers in Phase 5–6 combine these signals to achieve substantially higher discrimination (Table 16).

12.2 Feature Distributions: Correct vs. Hallucinations

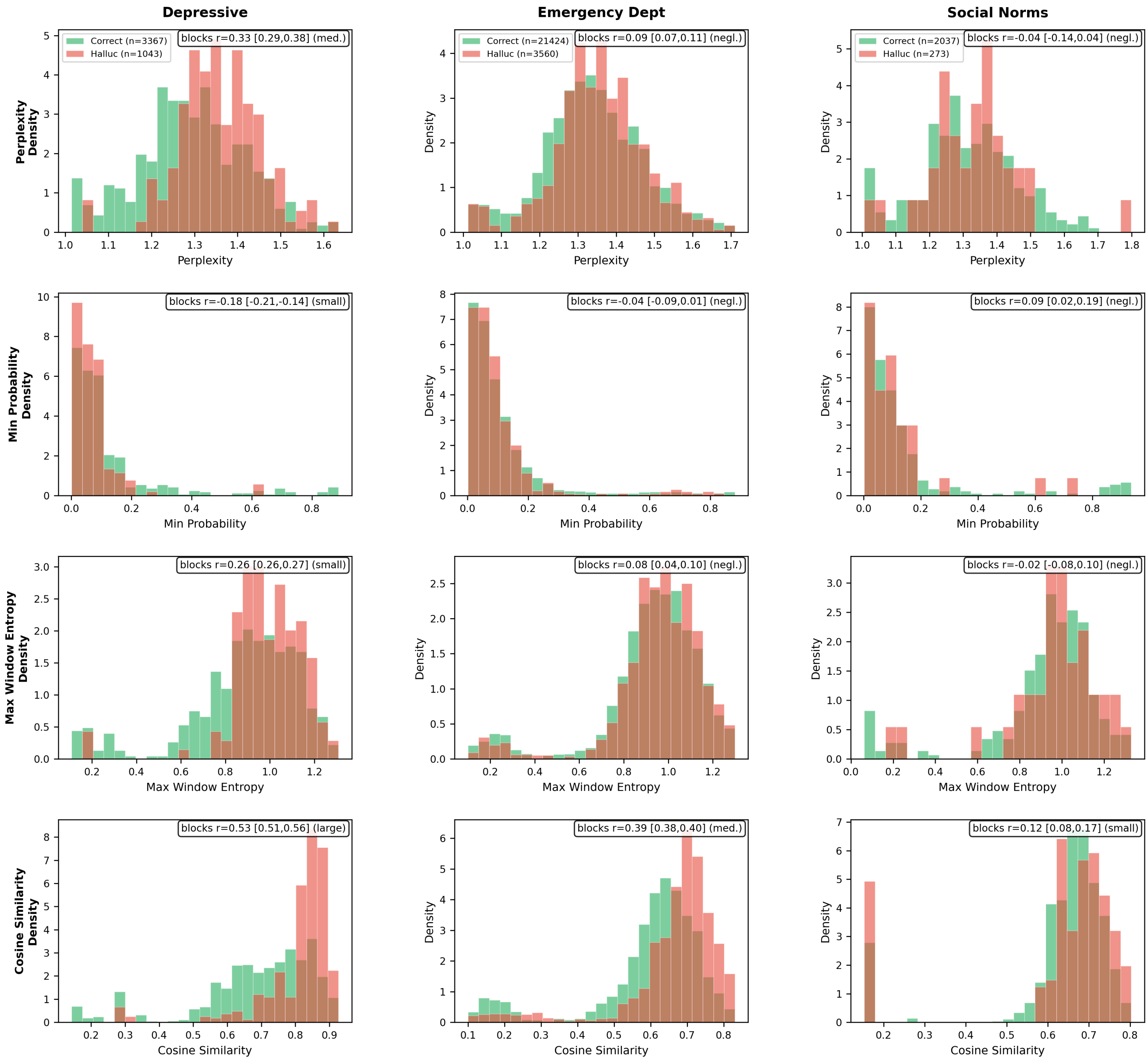

Figure 10: Generator UQ metric distributions comparing correct edges (green) vs. hallucinations (red) for each CLD (99th percentile). Rows show generator metrics; columns show CLDs. Subplot annotations report block-level rank-biserial effect sizes r aggregated across runs within each CLD (mean [min, max] over the 3 (CLD×run) blocks), rather than edge-level p-values.

To contextualize these distributions, we first assess whether the UQ metrics are approximately normal—a key assumption for many common parametric comparisons. Table 15 reports Shapiro–Wilk normality tests (overall, hallucinations only, and correct edges only) for each generator-based metric. All tests reject normality at $\alpha = 0.05$, consistent with the pronounced skewness and heavy tails observed in Figure 10. Consequently, edge-level distribution comparisons are treated as descriptive/diagnostic. Statistical inference for RQ2 is performed at the block level (CLD×run, $N = 9$) using one-sample t-tests and t-based 95% CIs on block-mean AUCs (Table 14).

Table 15: Shapiro-Wilk Normality Tests for UQ Metric Distributions

UQ Metric	**Group**	**N**	**Skew**	**Kurt**	**W**	**p-value**	**Normal?**
Perplexity	Overall	31,704†	+16.30	+263.7	0.037	$< 10^{-50}$	No
	Halluc	4,876	-0.07	+1.5	0.973	1.58×10^{-29}	No
	Correct	26,828†	+14.98	+222.4	0.035	$< 10^{-50}$	No
Min Prob	Overall	31,704†	+3.17	+10.5	0.608	$< 10^{-50}$	No
	Halluc	4,876	+3.71	+14.8	0.542	$< 10^{-50}$	No
	Correct	26,828†	+3.09	+10.0	0.598	$< 10^{-50}$	No
Max Window Entropy	Overall	31,704†	-1.60	+2.9	0.853	$< 10^{-50}$	No
	Halluc	4,876	-1.88	+4.9	0.831	$< 10^{-50}$	No
	Correct	26,828†	-1.56	+2.6	0.850	$< 10^{-50}$	No
Cosine Similarity	Overall	16,507†	-1.54	+2.4	0.842	$< 10^{-50}$	No
	Halluc	2,514	-1.87	+4.5	0.827	5.63×10^{-46}	No
	Correct	13,993†	-1.56	+2.4	0.836	$< 10^{-50}$	No

Note. Shapiro-Wilk test for normality (null hypothesis: data is normally distributed). †Subsampled to n = 5,000 due to Shapiro-Wilk sample size limit. Skew = skewness (0 = symmetric; positive = right-tailed); Kurt = excess kurtosis (0 = normal; positive = heavy-tailed). All metrics reject normality at $\alpha = 0.05$, justifying non-parametric methods (Mann-Whitney U, AUC-ROC).

Interpretation: The normality diagnostics confirm that the generator-based UQ metrics are strongly non-Gaussian at the edge level (both overall and within hallucination/correct strata). This cautions against over-interpreting visual separation alone and motivates non-parametric, distribution-robust summaries for within-CLD diagnostics. For inferential claims about RQ2 (above-chance discrimination across independent generation scenarios), we rely on block-level aggregation (CLD$\times$run) rather than edge-level p-values to avoid pseudo-replication.

Figure 10 compares LLM generator uncertainty metric distributions for correct edges (green) versus hallucinations (red) across the three CLDs (99th percentile clipped; Mann–Whitney U effect sizes r shown per subplot). Columns show different CLDs; rows show different metrics. Overall, the figures show unclear separation between hallucinated and non-hallucinated edges visually, though statistical tests show differences.

Gen perplexity distributions seem to shift slightly to the left for correct edges in depressive, and have less positive kurtosis than hallucinations in emergency department and social norms, although p-values show significance in depressive and emergency department but not in social norms.

Minimum token probability exhibits the most consistent separation (depressive p¡0.0001; emergency_department p=0.0004; social_norms p=0.0806). Valid edges tend to have a tail of higher minimum probability across all CLDs, although most of the probability mass of the correct distribution is also around the low-probability area.

Maximum window entropy shows significance in emergency department and depressive but not in social norms, with hallucinations skewing toward higher local entropy peaks in the depressive and emergency_department CLDs (both p¡0.0001), while no separation is observed for social_norms (max window entropy p=0.729), where the distribution appears slightly bimodal. Finally, cosine similarity to the retrieved context (available only for citation judging runs) differs significantly in all three CLDs (depressive and emergency_department p¡0.0001; social_norms p=0.0365), with hallucinations tending toward higher similarity, suggesting that semantic proximity to retrieved context does not guarantee causal correctness and may reflect topical alignment rather than faithful edge generation.

Overall, evidence suggests that these metrics carry some signal, but this seems to be domain dependent, is uneven and subtle, and does not clearly visually separate the data. Also, because of the relatively large sample size, significant effects are easily observed even if the effect is small, so do not necessarily denote useful effect sizes.

Even if no single metric yields a clear univariate threshold, the metrics may still be jointly informative: multivariate models can capture interactions and non-linear combinations that define a decision boundary not evident in marginal distributions. Therefore, we next train an ensemble classifier using all generator metrics jointly.

12.3 Recursive Feature Elimination Analysis

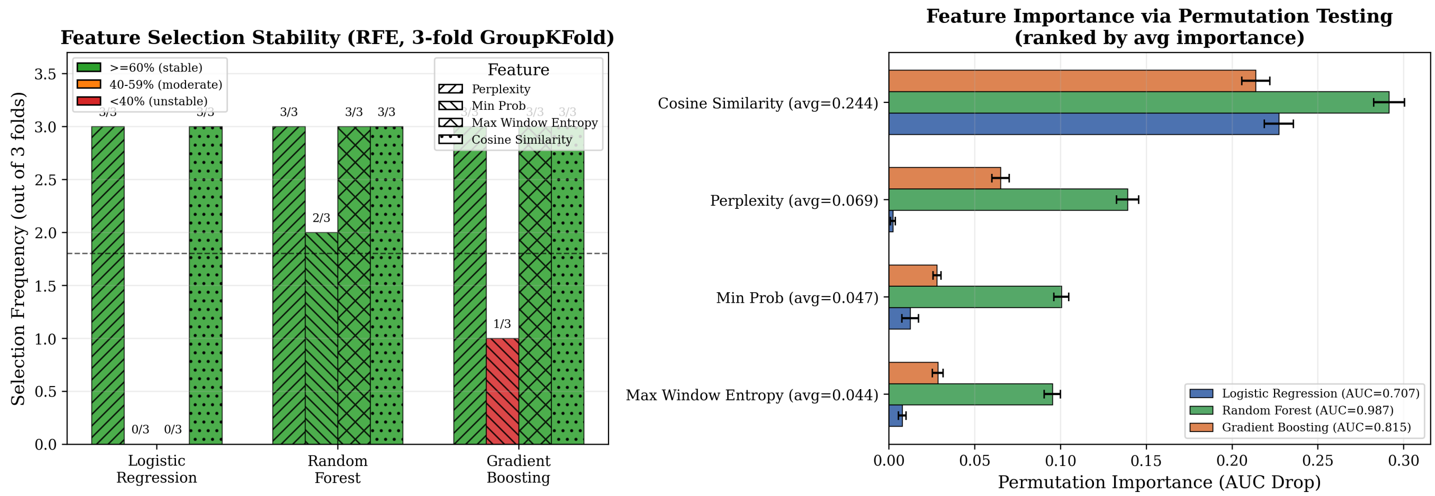

Figure 11: RFE analysis across classifiers with 3-fold block-level GroupKFold (blocks = CLD×run). **Left:** Feature selection stability with color-coded thresholds (green ≥60% = stable, orange 40–59%, red <40% = unstable); labels show selection frequency. **Right:** Permutation importance (AUC drop when feature is shuffled) for each classifier; error bars show standard deviation across 30 permutations.

Figure 11 reveals substantial differences in feature selection and importance across classifiers. Gen Cosine Similarity is the most important feature for all classifiers (permutation importance 0.21–0.29), consistent with its stable selection across all models. For Logistic Regression (AUC = 0.704), Gen Min Prob, Gen Max Window Entropy, and Gen Perplexity show near-zero permutation importance, explaining why RFE never selects them (0/3 folds). In contrast, Random Forest (AUC = 0.862) shows high permutation importance for all 4 features (0.10–0.29), justifying its selection of all features. However, Random Forest's superior in-distribution performance comes at the cost of poor cross-domain generalization (Table 16), suggesting that the additional features capture CLD-specific patterns rather than generalizable hallucination signals.

Having identified which features contribute to classification, we now evaluate ensemble classifiers using a cross-validation framework designed to detect overfitting and assess cross-CLD generalization.

12.4 Ensemble Classification Results

Ensemble classifiers require a consistent feature vector per edge. Therefore, Phases 4–6 are computed on the subset of edges for which all four UQ metrics are available (i.e., citation-judging runs only), yielding 16,507 edges across 35 experiment files—a reduction from the 31,704 edges across 63 files used in the single-metric analysis.

Table 16 presents a cross-phase validation analysis designed to detect overfitting. Each phase uses a progressively stricter validation strategy: Phase 4 tests feature selection on all pooled data; Phase 5 evaluates on held-out samples from the same distribution (all CLDs); Phase 6 tests generalization to entirely unseen CLDs (leave-one-CLD-out cross-validation). Performance drops between phases indicate the degree to which models exploit CLD-specific patterns rather than learning generalizable hallucination signals.

Table 16: Ensemble Classifier Performance Across Validation Phases (Block-Level Cross-Validation)

Classifier	**Phase 4: RFE CV AUC [min, max]**	**Phase 5: CV AUC [min, max]**	**Phase 5: Test AUC**	**Phase 6: Cross-CLD AUC [min, max]**	**Performance Drop (5→6)**
Logistic Regression	0.704 [0.670, 0.726]	0.695 [0.670, 0.732]	0.702	0.664 [0.562, 0.753]	0.039
Random Forest	0.862 [0.800, 0.985]	0.832 [0.701, 0.995]	0.809	0.629 [0.579, 0.712]	0.179
Gradient Boosting	0.766 [0.756, 0.775]	0.743 [0.725, 0.777]	0.765	0.645 [0.530, 0.767]	0.120
Neural Network	N/A	0.709 [0.693, 0.724]	0.739	0.648 [0.569, 0.738]	0.092

Note. Block-level cross-validation using 3-fold GroupKFold with blocks = (CLD×run, N = 9). Entire blocks stay together in train OR test to prevent pseudo-replication. **Phase 4 (RFE CV AUC):** 3-fold block-level CV on all pooled edges; measures feature selection performance. **Phase 5 (CV AUC):** 3-fold block-level CV on training blocks (∼67% of blocks); measures in-distribution performance. **Phase 5 (Test AUC):** Single evaluation on held-out test blocks (∼33% of blocks); validates generalization within same distribution. **Phase 6 (Cross-CLD AUC):** Leave-one-CLD-out CV—train on 2 CLDs, test on held-out 3rd CLD, averaged across all 3 CLDs; measures cross-domain generalization. Phase 4 evaluates all 4 UQ metrics via RFE for each classifier; the best-performing classifier (Random Forest) selected all 4 features, which are then used in Phases 5–6. **Uncertainty:** All phases show mean [min, max] across CV folds or CLDs, per the uncertainty reporting rule (Methods Section **??**). For $n = 3$ (Phases 4–5) or $n = 3$ (Phase 6), min-max ranges are more appropriate than t-distribution CIs. Performance Drop = Phase 5 Test AUC − Phase 6 Mean AUC.

Table 16 shows cross-phase validation results using block-level GroupKFold (blocks = CLD×run, N = 9). Random Forest achieves the highest in-distribution performance (Phase 5 Test AUC = 0.809) but exhibits substantial degradation under leave-one-CLD-out evaluation (Phase 6 mean AUC = 0.629, drop = 0.179). Gradient Boosting shows similar overfitting (Phase 5 = 0.765, Phase 6 = 0.645, drop = 0.120). In contrast, Logistic Regression and the Neural Network show smaller cross-domain degradation (drops of 0.039 and 0.092) and similar Phase 6 mean AUC (0.664 vs 0.648; overlapping min–max ranges). The block-level CV prevents pseudo-replication from correlated edges within the same generation scenario, yielding more conservative but trustworthy performance estimates. Overall, these results suggest that simpler models generalize more robustly across CLDs, while higher-capacity tree ensembles are more prone to CLD-specific overfitting.

12.5 RQ2: Main Findings (Summary)

RQ2 asks whether generator-time uncertainty (UQ) metrics can predict hallucinated edges (FP/FN w.r.t. GT Lit) in CLD generation. Across the single-metric and ensemble analyses, we find:

1. **Single metrics carry only weak-to-moderate signal.** The strongest single metric is Gen Cosine Similarity (AUC = 0.671; r = 0.153; Table 14), while the remaining generator-time metrics provide weaker discrimination (AUC ≈ 0.55). Notably, cosine similarity exhibits a counterintuitive direction (higher similarity associated with hallucinations), consistent with topical overlap rather than causal faithfulness.
2. **High-capacity ensembles overfit across domains.** Random Forest achieves the highest in-distribution performance (Phase 5 Test AUC = 0.809) but degrades under leave-one-CLD-out evaluation (Phase 6 mean AUC = 0.629), indicating CLD-specific overfitting (Table 16).
3. **Simpler models generalize better, but remain limited.** Logistic Regression and the Neural Network show the smallest cross-domain performance drops (0.039 and 0.092) and comparable Phase 6 mean AUC (∼0.65–0.66; Table 16), implying some generalizable signal, but not strong enough to support reliable cross-domain hallucination detection in isolation.

Takeaway: Single generator UQ metrics provide a weak signal for hallucination detection. When used in an ensemble as input features for classifiers, they perform well in-distribution for some models (Gradient Boosting, Random Forest) but show limited cross-CLD generalization, reducing practical applicability. In this study, UQ metrics are better viewed as auxiliary features for triage/monitoring rather than as a standalone, domain-robust hallucination detector.

RQ3 Pivot Rationale. The original RQ3 design—using generator UQ metrics to selectively deploy the corrector—is not supported by our findings: RQ2 shows that UQ metrics do not generalize across CLDs (Phase 6 AUC ≈ 0.63–0.67), and RQ1b shows the corrector does not reliably improve GT Lit quality (macro ΔF1 ≈ −0.02). We therefore pivot to an alternative form of smart test-time compute: using the Mechanistic Correctness Judge score (which showed more reliable discrimination in

RQ1a GT Lit; Table 13) as the uncertainty signal, and **Deep Research** (DR)—automated multi-agent literature search—as the form of deploying more (expensive) test-time compute that can measurably validate causal claims.

This preserves the spirit of RQ3: a computationally cheap uncertainty signal triggers selective deployment of more expensive but more effective test-time compute. The RQ1a physics CLD validation (Section **??**) provides empirical motivation for DR: simple citation retrieval achieves 89.5% retrieval coverage on Ulemans CLD but yields low judge scores, while physics CLDs with lower retrieval (55%) achieve high approval when retrieved (82.4%). The bottleneck is not retrieval success but **evidence type**—standard search finds correlational literature in complex domains while physics literature contains definitive causal statements. DR addresses this by specifically targeting direct causal evidence (RCTs, experimental manipulations, meta-analyses) through iterative, multi-agent search that filters for study types establishing causation rather than correlation. Due to time constraints from the late pivot, we did not explore the DR design space as systematically as in RQ1a/RQ1b with a component-wise ablation study. The DR system uses multiple search queries in parallel, iterative evidence gathering, and a multi-agent pipeline for scoring, selection, and judgment (see Methods, Section **??**, for full system architecture).

Limitations of the exploratory design. The DR system has many degrees of freedom (query count, iteration depth, scoring thresholds, agent prompts, and components) that would ideally require a full ablation study with local and global sensitivity analysis. Additionally, searching for evidence *for* a claim introduces confirmation bias risk, and the system inherits publication bias from the underlying literature. These limitations are acknowledged; the following results should be interpreted as an exploratory pilot rather than a definitive evaluation.

13 RQ3: Deep Research Validation (Exploratory Pilot)

Methodological pivot. The original RQ3 (smart test-time compute using UQ metrics to selectively deploy correctors) was contingent on RQ2 generalization and RQ1b corrector efficacy—conditions not supported by our findings. We therefore repurposed RQ3 as an exploratory pilot of **Deep Research (DR)**: using automated literature search to *validate* (rather than correct) LLM-generated causal claims. This pilot is single-run and non-ablated; see Discussion (RQ3 pivot section) for rationale and limitations.

We deployed DR on 285 edges (TP + FP + FN from the RQ1a GT Lit labels across all three CLDs; TN omitted to reduce cost) and asked whether automated literature search can validate the LLM's causal *narrative* for each edge. The key metric is **detection rate**: the fraction of edges for which DR retrieves *direct causal evidence* (RCTs, experimental manipulations, or meta-analyses; not purely correlational studies). We additionally report DR's self-assessed confidence (1–10).

Table 17: Deep Research Detection Rate and Discrimination Analysis (n=285 edges)

Classification	**n**	**Detection Rate**	**Confidence**	**TP–X Gap**	**Cohen's h**	**p-value**
TP (Expert-accepted)	44	70.5%	6.59	—	—	—
FP (LLM hallucinations)	178	62.4%	6.64	8.1 pp	0.17	.38
FN (Expert-rejected)	63	42.9%	6.78	27.6 pp	0.56	.006**

Note. Detection rate = % edges with direct causal evidence found. Confidence = self-assessed (1–10 scale). TP/FP/FN from RQ1a GT Lit validation across all 3 CLDs.
TP–FP test. Fisher's exact $p = .38$; OR = 1.44 [95% CI: 0.70, 2.94]; Cohen's $h = 0.17$ [95% CI: -0.16, 0.50] (negligible); power = 17%.
TP–FN test. Fisher's exact $p = .006**$; Cohen's $h = 0.56$ [95% CI: 0.18, 0.95] (medium); power = 82%.
FP–FN test. Fisher's exact $p = .008**$; Cohen's $h = 0.39$ [95% CI: 0.11, 0.68] (small); power = 76%.
Multiple comparisons. Bonferroni correction for 3 pairwise tests. TP–FP $p_{\text{adj}} = 1.00$; TP–FN $p_{\text{adj}} = .02*$; FP–FN $p_{\text{adj}} = .02*$.

Table 17 shows that TP edges have a higher detection rate than FP edges, but the TP–FP gap (8.1 pp) is not statistically significant ($p = .38$; low power, effect-size CI includes zero). In contrast, detection differs substantially between TP and FN (27.6 pp; significant after correction; medium effect) and between FP and FN (19.5 pp; significant after correction; small effect).

Given these aggregate results, we next report per-CLD detection rates to assess whether DR's discriminative behavior is domain-dependent.

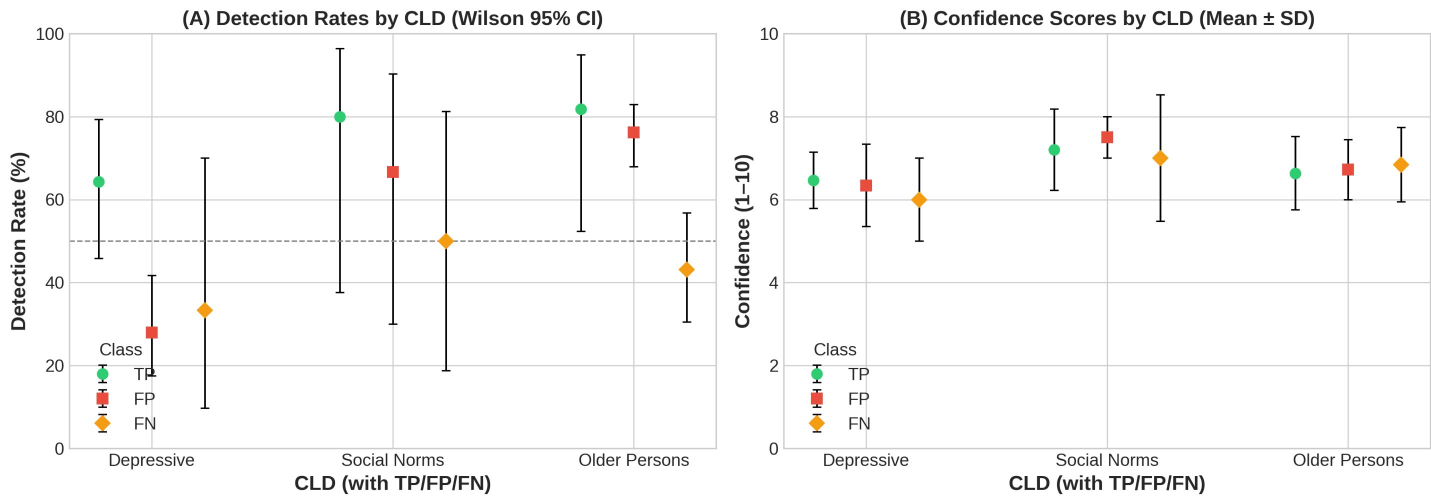

Figure 12: Deep Research results. **(A)** Detection rates by CLD and classification (TP/FP/FN) with Wilson 95% CIs. **(B)** Confidence scores by CLD and classification (mean ± SD).

Figure 12A shows substantial between-CLD variation in detection gaps between edge classes: Depressive exhibits meaningful discrimination (TP–FP gap = +36 pp), while Social Norms (+13 pp) and Older Persons (+6 pp) show weak discrimination; uncertainty is substantial for Social Norms due to small n. Figure 12B shows DR's self-reported confidence provides little discrimination between edge classes: means cluster tightly (range $<$ 1 point on a 10-point scale) with substantial overlap across TP/FP/FN within each domain.

In aggregate, DR does not significantly discriminate between valid and hallucinated (as defined as FP in GT Lit) causal claims (TP–FP gap = 8.1 pp; $p = .38$; Cohen's $h = 0.17$, negligible). Nonetheless, DR retrieves direct causal evidence for 62.4% of FP edges and 42.9% of FN edges. This is consistent with multiple explanations: (i) some "false positives" may reflect literature-supported relationships that experts excluded for scope/theoretical reasons, and/or (ii) DR may retrieve evidence that is not sufficiently specific or causal for the exact claim—a motivation for the human validation causal evidence applicability analysis below. The FP–FN gap (+19.5 pp, $p = .008^{**}$, $p_{\text{adj}} = .024^{*}$) survives correction: FP edges have higher detection than FN edges.

13.1 Per-CLD Detection Rate Analysis

Table 18: Per-CLD Pairwise Discrimination Tests (Fisher's Exact, Bonferroni $m = 3$)

CLD	Comparison	Δpp	Cohen's h	Interpretation	p	p_{adj}
Depressive	TP–FP	+36.3	0.75	medium	.004**	.01*
	TP–FN	+31.0	0.63	medium	.20	.61
	FP–FN	-5.3	-0.12	negligible	1.00	1.00
Social Norms	TP–FP	+13.3	0.30	small	1.00	1.00
	TP–FN	+30.0	0.64	medium	.55	1.00
	FP–FN	+16.7	0.34	small	1.00	1.00
Older Persons	TP–FP	+5.6	0.14	negligible	1.00	1.00
	TP–FN	+38.7	0.83	large	.04*	.13
	FP–FN	+33.1	0.69	medium	¡.001***	¡.001***

Note. All pairwise comparisons (TP–FP, TP–FN, FP–FN) within each CLD using Fisher's exact test; Bonferroni correction applied within each CLD ($m = 3$, $\alpha_{\text{adj}} = 0.017$). Effect size interpretation: $|h| < 0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, > 0.8 large. Detection rates and confidence distributions shown in Figure 12.

Table 18 reveals substantial domain variation in DR's TP–FP discriminative ability, relatively smaller variation in TP–FN, and substantial variation in the FP–FN gap. In the Depressive CLD, DR clearly distinguishes valid from hallucinated edges

(TP–FP: +36.3 pp, $h = 0.75$, $p_{\text{adj}} = .01$). The Social Norms CLD shows a smaller, non-significant gap (TP–FP: +13.3 pp, $h = 0.30$). In contrast, the Older Persons CLD shows negligible TP–FP discrimination (+5.6 pp, $h = 0.14$), but a significant FP–FN gap (+33.1 pp, $h = 0.69$, $p_{\text{adj}} < .001$), indicating DR validates most edges—including false positives—while consistently missing edges the LLM failed to generate.

13.2 Exploratory: Ground Truth Enrichment Analysis

Given that 62.4% of FP edges have `direct_causal_found=True` under DR, we explore an optimistic upper bound: *if* DR-validated FPs were treated as true causal relationships, how would generation performance change? This assumption is intentionally strong and is used only to bound the potential impact; the human-validation analysis below provides a more conservative calibration.

Table 19: LLM Generation Metrics: Two Enrichment Scenarios (Aggregate and Per-CLD)

	Original	Scenario 1: Enriched GT		Scenario 2: LLM+DR System	
Dataset	**F1**	**F1**	**FPs → TP**	**F1**	**FNs found**
Aggregate	0.267	0.705 (+163%)	111/178 (62%)	0.779 (+191%)	27/63 (43%)
Depressive	0.500	0.667 (+33%)	14/50 (28%)	0.687 (+37%)	2/6 (33%)
Social Norms	0.455	0.692 (+52%)	4/6 (67%)	0.828 (+82%)	3/6 (50%)
Older Persons	0.113	0.722 (+540%)	93/122 (76%)	0.813 (+621%)	22/51 (43%)

Scenario 1 (Enriched GT): Promotes FP edges with DR evidence to TP. Tests: "What if experts missed valid relationships?"
Scenario 2 (LLM+DR System): Scenario 1 + DR discovers FN edges with literature support. Tests: "What could a complete LLM+DR pipeline achieve?"
Aggregate confusion matrices: Original: TP=44, FP=178, FN=63. Scenario 1: TP=155, FP=67, FN=63. Scenario 2: TP=182, FP=67, FN=36.

Of the 63 FN edges (expert-included, LLM-missed), 27 (42.9%) have DR causal evidence. Under Scenario 2, a complete LLM+DR pipeline would recover these edges as positive relationships, reducing FN from 63 to 36. The remaining 36 FN edges (57.1%) lack searchable evidence under our DR settings, consistent with tacit domain knowledge or evidence types not captured by the retrieved literature. Scenario 1 quantifies the impact of promoting FP→TP (F1: 0.267 → 0.705), while Scenario 2 estimates an upper bound when additionally recovering FN edges with DR evidence (F1: 0.267 → 0.779). The 36 residual FNs represent expert knowledge that neither LLM generation nor literature search recovers in this setting.

Human validation sampling and labels. We stratified-sampled 50 edges from the Deep Research output restricted to `direct_causal_found=True` and manually labeled each edge with a categorical verdict (`human_verdict`∈{`causal`, `different_vars, causal`, `Not relevant / non-causal`}) and a human-assigned evidence applicability score `Fully applicable`$\in [0, 1]$. The sampling protocol and applicability rubric are defined in Methods (Section **??**).

Human validation evidence verdicts. Table 20 summarizes the distribution of human verdicts by edge class. All TP edges are supported by direct causal evidence, but a majority of FP edges also have causal evidence (either exact-match `causal` or `different_vars, causal`). This corroborates that the primary failure mode is *variable mismatch* (causal evidence for related constructs/measures), rather than outright non-causal evidence.

Table 20: Human Validation: Evidence Verdict Distribution by Edge Classification ($n = 50$)

Verdict	**TP** ($n = 9$)	**FP** ($n = 32$)	**FN** ($n = 9$)	**Total**
`causal`	9 (100.0%)	21 (65.6%)	5 (55.6%)	35 (70.0%)
`different_vars, causal`	0 (0.0%)	8 (25.0%)	2 (22.2%)	10 (20.0%)
`Not relevant / non-causal`	0 (0.0%)	3 (9.4%)	2 (22.2%)	5 (10.0%)
Any causal	9 (100.0%)	29 (90.6%)	7 (77.8%)	45 (90.0%)

Note. `causal` = exact match to edge claim; `different_vars, causal` = causal evidence for related but not identical variables/constructs; `Not relevant / non-causal` = evidence does not support a causal relationship. "Any causal" = `causal` ∪ `different_vars, causal`.

Table 21: Applicability Score Rubric for Human Validation

Score	Description
1.0	Causal evidence for exact variable constructs and measures
0.9	Causal evidence, measure differing for one variable
0.8	Causal evidence, measure differing for two variables
0.7	Causal evidence, one variable differing but believably related
0.6	Causal evidence, two variables differing but believably related
0.5	Non-causal evidence for variables in question
$\leq$0.4	Progressively less related evidence

Applicability-based calibration (thesis-ready filtering). Because DR's binary `direct_causal_found=True` flag only indicates that *some* direct causal evidence was retrieved—and often fails via variable/operationalization mismatch—we calibrated it using the human `Fully applicable`$\in [0,1]$ score. For thresholds $t \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, we computed the pooled coverage–correctness tradeoff (Figure 13/Table 22).

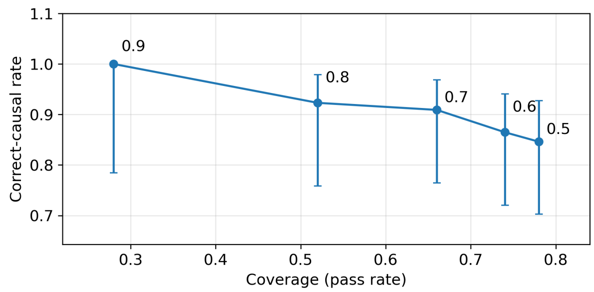

Figure 13: Human validation tradeoff curve (pooled; Wilson 95% CI). Each point corresponds to an applicability threshold $t \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ applied to the validated sample ($n = 50$). Let $x(t)$ be *coverage* $= \Pr(\texttt{Fully\_applicable} \geq t)$ and $y(t)$ be *correct-causal rate* $= \Pr(\texttt{human\_verdict} = \texttt{causal} \mid \texttt{Fully\_applicable} \geq t)$. Points are annotated by t. Selected threshold: $t = 0.7$.

Table 22: Human validation tradeoff curve (pooled): coverage vs. correct-causal rate

Threshold	Coverage	Correct-causal rate	n pass
0.5	0.780	0.846	39
0.6	0.740	0.865	37
0.7	**0.660**	**0.909**	**33**
0.8	0.520	0.923	26
0.9	0.280	1.000	14

Coverage = fraction of validated edges with applicability $\geq t$. Correct-causal rate = fraction of passing edges labeled `causal`.
Bold row indicates selected threshold $t = 0.7$.

We then selected $t = 0.7$ via the script's elbow/gradient decision rule and used the human-validated pass rates to extrapolate expected FP promotions and FN discoveries to the full DR dataset (Table 23).

Table 23: LLM Generation Metrics with Human-Validated Ground Truth Enrichment (Aggregate, extrapolated)

Scenario	Precision	Recall	F1
Original (expert GT)	0.198	0.411	0.267
Scenario 1: Enriched GT (FP$\rightarrow$TP)	0.459	0.618	0.527
Scenario 2: LLM+DR System (+FN found)	0.494	0.709	0.582
Conservative Scenario 1	0.396	0.583	0.472
Conservative Scenario 2	0.415	0.629	0.500

Threshold selected by elbow/gradient rule (Option B) on validation tradeoff curve: $t = 0.7$. FP promotions (point / conservative): 58 / 44. FN discoveries (point / conservative): 15 / 7. Counts are extrapolated from a stratified human-validation sample and should be interpreted as estimates.

Critical Caveats: Both scenarios are **exploratory and preliminary**. Key limitations: (1) DR may validate edges as being directly causal irrespective of other CLD specific context such as mediators. (2) LLM-based DR validating LLM-generated edges risks circularity; (3) The review of promoted/discovered edges was solely done by the author of this thesis and not a domain expert and has no inter rater reliability reporting; (4) **True Negatives not analyzed**—DR covered only 20.4% of the edge space (285 of 1,394 possible edges); the 1,109 TN edges were not analyzed due to cost (\$0.87/edge, \$269 for all TNs). These metrics represent **upper bounds** on achievable performance, pending domain expert validation. Dataset available for community verification (see Data Availability).

13.3 Smart-Trigger DR Deployment Estimate

The enrichment analysis above applies DR to *all* edges. In practice, DR is expensive ($0.87/edge), motivating a **smart-trigger** deployment that runs DR selectively. We estimate expected F1 gains if DR is triggered only for edges where the most reliable RQ1a Ground Truth Correctness judge (Mechanistic prompt) assigns a low aggregate score (≤ 0.5), indicating uncertainty about the edge (as opposed to the original RQ3 intention of using RQ2 UQ metrics as the trigger).

Procedure. We parsed per-edge Mechanistic judge scores from the 9 judged workbooks (3 CLDs $\times$ 3 runs) and computed the mean score per edge. We then joined these to the 285 DR edges and marked edges as *triggered* if their mean score ≤ 0.5. Of the triggered edges, we counted how many had `direct_causal_found=True` in the DR oracle. Finally, we applied the human-validated calibration rates at $t = 0.7$ (FP: 89.5%, FN: 83.3%) to estimate expected promotions/discoveries.

Results. Of 285 DR edges, 142 triggered (50%). Among triggered edges with DR evidence: 59 FP edges and 9 FN edges had `direct_causal_found=True`. Applying calibration:

Table 24: Smart-trigger Deep Research enrichment estimates. Trigger: Mechanistic GT correctness score ≤ 0.5. Calibration: human-validated applicability ≥ 0.7.

Scenario	Precision	Recall	F1 (95% CI)
Original	0.198	0.411	0.267
Enriched GT (+52.8 FP→TP)	0.436	0.606	0.507 [0.457, 0.524]
LLM+DR (+7.5 FN discovered)	0.454	0.653	0.536 [0.474, 0.557]

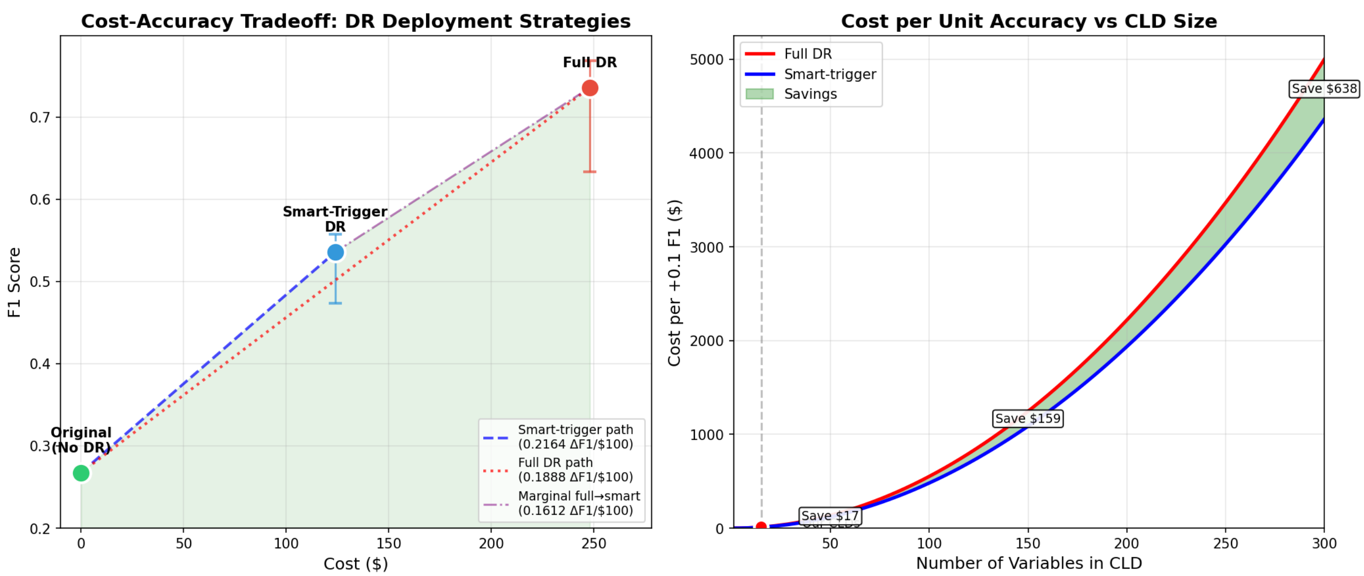

Figure 14: Cost–accuracy tradeoff and cost-per-unit-accuracy scaling for Deep Research (DR) deployment. Left: observed cost vs. expected F1 for always-on DR vs. smart-trigger DR (judge-triggered deployment). Right: extrapolated cost per +0.1 F1 improvement as CLD size increases under a sparse-edge growth assumption, highlighting the widening cost gap.

Note. Figure 14 shows two-sided uncertainty bands (Wilson 95% bounds) propagated from the human-calibrated promotion/discovery rates (see Methods, Sections **??** and **??**). For interpretation, we prefer the conservative estimate obtained by using the Wilson 95% lower bound, because it reduces the risk of overstating gains due to sampling variability in the human validation rates. While this lower bound addresses sampling uncertainty in the calibration step, it does not account for other sources of uncertainty (e.g., generalization of applicability to non-validated edges, edge dependence, domain shift, and not considering CLD mediators). Further research should assess how the estimated gains change when these assumptions are relaxed.

We evaluated whether a selective test-time DR deployment policy could improve cost-efficiency relative to always-on DR, using the RQ1a Ground Truth Correctness judge (Mechanistic prompt) as a trigger (mean aggregate score ≤ 0.5). On the 285-edge RQ3 dataset, 142 edges (50%) triggered; among these, DR reported `direct_causal_found=True` for 59 FP and 9 FN edges. Using the human-validated calibration at applicability ≥ 0.7 (FP: 17/19 causal; FN: 5/6 causal; Wilson 95% CIs reported above), the smart-trigger policy yields an estimated ΔF1 of 0.268 at $\sim$\$124 total compute (judge on all edges +

DR on triggered edges), compared to ΔF1 of +0.468 at ∼$248 for full DR.

Under these assumptions, smart-trigger achieves higher ΔF1 per dollar (0.216 vs. 0.189 ΔF1 per $100; ∼15% relative improvement) while recovering ∼57% of the full-deployment F1 gain at ∼50% of the DR cost. The scaling panel should be interpreted as a cost extrapolation, not a validated performance claim: it assumes the observed trigger rate and per-edge applicability distribution generalize to other edges and other CLDs. Under this stylized model, the absolute dollar gap between full and smart-trigger deployment increases with CLD size due to quadratic scaling of the edge space, while the efficiency ratio remains approximately constant.

14 RQ3: Main Findings (Summary)

RQ3 asks whether automated literature search (Deep Research; DR) can validate LLM-generated causal edges and improve end-to-end CLD quality. As an exploratory pilot (single-run, non-ablated), we find:

1. **DR detection is not a reliable FP filter in aggregate.** Across 285 TP/FP/FN edges, DR retrieves `direct_causal_found=T` for many FP edges (62.4%), yielding a small, non-significant TP–FP detection gap (8.1 pp; $p = .38$; Cohen's $h = 0.17$; Table 17). DR does, however, distinguish FN from both TP and FP (TP–FN and FP–FN gaps significant after correction; Figure 12).
2. **Utility is domain-dependent and DR confidence is weakly informative.** Per-CLD results show strong TP–FP discrimination in Depressive but weak discrimination in Social Norms and Older Persons (Table 18; Figure 12). DR's self-reported confidence shows substantial overlap across TP/FP/FN classes within domains (Figure 12B), limiting its value as a standalone accept/reject criterion.
3. **Enrichment effects can be large if gains extrapolated from the human validation sample hold.** A naive upper bound (treat all DR-positive edges with `direct_causal_found=True` as correct) yields very large gains (Scenario 1/2; Table 19). However, DR sometimes finds causal evidence for highly related, but not directly equivalent, concepts or using different measures. Human validation shows that the dominant failure mode is *variable mismatch*—causal evidence for different constructs than the edge's stated variables. To form a more realistic estimate based on the human validation sample, an optimal trade-off between evidence quality and coverage can be found by applying a threshold on the pooled tradeoff curve ($t = 0.7$; Figure 13), which yields more calibrated enrichment estimates than the naive "any causal evidence found by DR is valid" assumption: F1 improves from 0.267 (expert GT) to 0.527 (Scenario 1) and 0.582 (Scenario 2), with conservative estimates of 0.472 and 0.500 (Table 23). Based on the quality of causal evidence observed in the human validation sample, the human-calibrated enrichment extrapolation seems mostly reasonable, but it relies on strong assumptions: (i) the within-CLD edge-class applicability distribution in the validated sample generalizes to the remaining (out-of-sample) edges for that CLD, and (ii) edges are evaluated largely in isolation rather than in the full CLD causal context (e.g., mediators/confounders) that could disqualify a relationship from being directly causal.
4. **Smart-trigger DR slightly improves cost-efficiency (exploratory).** Triggering DR only on edges flagged as uncertain by the strongest correctness judge (Mechanistic; score ≤ 0.5) recovers a notable fraction of the full-deployment DR enrichment gains at roughly half the DR cost, with an estimated ΔF1 of 0.268 after human-calibrated applicability filtering (Figure 14). This provides a proof of concept that DR efficiency can be improved by deploying DR selectively using lower-compute uncertainty estimates.

Takeaway: In this pilot, DR is best interpreted as a high-cost, domain-dependent *enrichment signal* that requires human-calibrated applicability thresholds; it is not a drop-in validator that cleanly separates expert-true edges from LLM false positives across domains. It can, however, yield direct causal evidence for a majority of edges in our CLD ground truth sample (under strong assumptions) that, if validated, could be used to improve CLD quality, with smart triggering used to partially offset DR's computational cost and achieve higher accuracy per dollar spent.

15 Effect Size Summary

Table 25 summarizes the standardized effect sizes observed across all research questions, linking each finding to its stated hypothesis, quantitative formulation, statistical test, and conclusion. This provides a unified view of which null hypotheses were rejected and the magnitude of effects detected.

Table 25: Effect Size Summary: Hypothesis Operationalization, Statistical Tests, and Conclusions

RQ	Hypothesis	H_0 / H_1	Test	Effect	Value	p	Conclusion (decision; effect)
RQ1a: LLM-as-a-Judge Validation							
1a	LLM-human agreement	H_0: $\kappa = 0$ vs H_1: $\kappa > 0$	Cohen's κ (n=72)	κ	0.618	—	H_0 rejected; substantial
1a	Prompt affects F1 (Corr. Citation)	H_0: equal medians H_1: $\geq$1 differs	Friedman (9 blocks)	W	0.28	.058	H_0 not rejected; weak
	Prompt affects F1 (Corr. Correctness)			W	0.70	.001**	H_0 rejected; strong
	Prompt affects F1 (GT Citation)			W	0.30	.042*	H_0 rejected; moderate
	Prompt affects F1 (GT Correctness)			W	0.48	.013*	H_0 rejected; moderate
RQ1b: LLM-as-a-Corrector Evaluation							
1b	Corrector efficacy (Synthetic)	H_0: med(ΔF1)=0 H_1: med(ΔF1)$\neq$0	Wilcoxon signed-rank (9 blocks)	d	+1.18	.031*	H_0 rejected; large
	Corrector efficacy (Ground Truth)	H_0: med(ΔF1)=0 H_1: med(ΔF1)$\neq$0	Wilcoxon signed-rank (9 blocks)	d	−0.32	.301	H_0 not rejected; medium
RQ2: Context-Insensitive Hallucination Detection							
2	UQ discriminates hallucinations	H_0: AUC=0.5 H_1: AUC$\neq$0.5	One-sample t-test (9 blocks)	AUC	0.672	<.001***	H_0 rejected; moderate
	Cross-CLD generalization	—	Descriptive (Phase 5→6)	ΔAUC	0.18	—	Poor generalization
RQ3: Deep Research Literature Validation[*]							
3	DR discriminates (TP–FP)	H_0: $p_{\mathrm{TP}} = p_{\mathrm{FP}}$ H_1: $p_{\mathrm{TP}} \neq p_{\mathrm{FP}}$	Fisher's exact (Bonf. m=3)	h	0.17	0.382	H_0 not rejected; negl.
	DR discriminates (TP–FN)	H_0: $p_{\mathrm{TP}} = p_{\mathrm{FN}}$ H_1: $p_{\mathrm{TP}} \neq p_{\mathrm{FN}}$	Fisher's exact (Bonf. m=3)	h	0.56	0.006	H_0 rejected; medium
	DR discriminates (FP–FN)	H_0: $p_{\mathrm{FP}} = p_{\mathrm{FN}}$ H_1: $p_{\mathrm{FP}} \neq p_{\mathrm{FN}}$	Fisher's exact (Bonf. m=3)	h	0.39	0.008	H_0 rejected; small
RQ3: Enrichment Analysis (Exploratory)[*]							
	Enrichment baseline	—	Descriptive	F1	0.267	—	Baseline
	Scenario 1: Enriched GT	—	Descriptive	ΔF1	+0.438	—	Exploratory
	Scenario 2: LLM+DR	—	Descriptive	ΔF1	+0.512	—	Exploratory
	Human-val. Scenario 1	—	Extrapolation	ΔF1	+0.260	—	Exploratory
	Human-val. Scenario 2	—	Extrapolation	ΔF1	+0.315	—	Exploratory
	Smart compute	—	Descriptive	ΔF1/\$100	+0.027	—	Exploratory

Effect size interpretation. κ: Landis-Koch (0.61–0.80 = substantial). W: <0.3 weak, 0.3–0.5 moderate, >0.5 strong. d: <0.2 small, 0.2–0.8 medium, >0.8 large. r: <0.2 weak, 0.2–0.5 moderate, >0.5 strong. h: <0.2 negligible, 0.2–0.5 small, 0.5–0.8 medium, >0.8 large.
Significance. * p < .05, ** p < .01, *** p < .001. RQ3 p-values are Bonferroni-adjusted ($\alpha_{\mathrm{adj}} = 0.0167$).
[*] *RQ3 exploratory.* Enrichment estimates are upper-bound/extrapolated; not independently validated ground truth.
RQ1a prompt tests. Friedman with (CLD×run) blocking; W = Kendall's concordance. *RQ1b efficacy.* One-sample Wilcoxon signed-rank on 9 blocks.
RQ2 discrimination. AUC = Area Under Curve; One-sample t-test on 9 block means (CLD×run) vs 0.5.

In summary, Table 25 shows: **RQ1a**—LLM-human agreement is substantial (κ=0.618, H_0 rejected); prompt effects on citation judging are weak (W=0.28–0.30) while correctness judging shows strong prompt sensitivity (W=0.70, H_0 rejected). **RQ1b**—corrector efficacy is confirmed for synthetic corruptions (H_0 rejected, d=+1.18, large effect) but not for ground truth (H_0 not rejected, d=−0.32, medium effect). **RQ2**—UQ metrics show statistically significant but weak discrimination (H_0 rejected, r=0.15) with poor cross-CLD generalization (ΔAUC=0.18). **RQ3**—no TP–FP discrimination (H_0 not rejected, h=0.17, negligible), but DR distinguishes TP and FP from FN (TP–FN: H_0 rejected, h=0.56, medium; FP–FN: H_0 rejected, h=0.39, small; both survive Bonferroni correction). Exploratory enrichment analyses indicate large aggregate improvements (ΔF1=+0.438 to +0.512), with human-validated extrapolations remaining substantial (ΔF1=+0.260 to +0.315).

16 Computational Scaling Analysis

16.1 Time and Cost Scaling

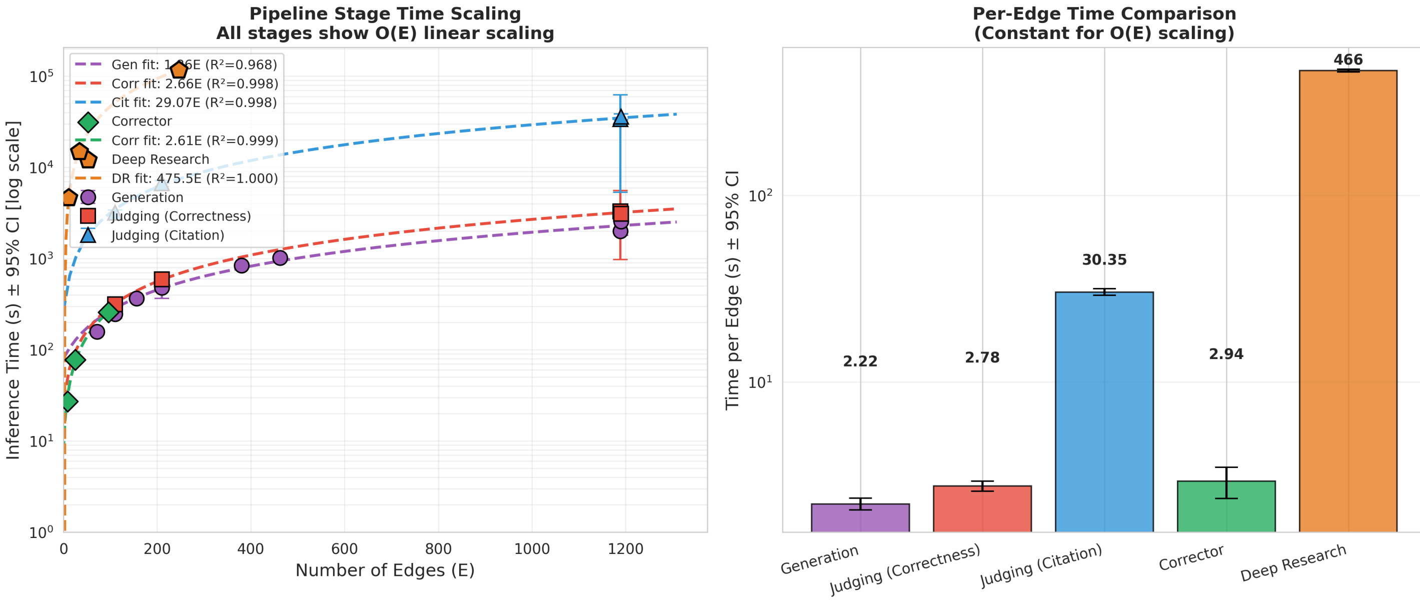

Figure 15: Generation vs. judging scaling comparison (mean ± SD). **Left:** Time scaling vs. number of edges. **Right:** Per-edge time comparison across pipeline stages. Error bars show SD.

Figure 15 shows that both generation and correctness judging exhibit $O(E)$ linear scaling with similar slopes ($\sim$2 s/edge). Generation (2.2 ± 0.5 s/edge) and correctness judging (2.8 ± 0.2 s/edge) have comparable costs, while citation judging (30.1 ± 1.4 s/edge) dominates the pipeline.

16.2 Token Cost Analysis

Table 26: Computational Efficiency Summary by Experiment

Experiment	**Model**	**Files**	**Edges**	**Tok/Edge**	**¢/Edge**	**Total ($)**
RQ1a Judge Experiments (actual)						
Corruption Citation	5m	36	18,120	42,992	1.37	247.67
Ground Truth Citation	4.1	36	18,117	43,933	9.22	1669.69
Corruption Correctness	4.1	27	13,590	480	0.16	22.28
Ground Truth Correctness	4.1	27	13,587	470	0.16	21.34
Human Validation	4.1	3	260	42,639	8.70	22.61
RQ1b Corrector Experiments (estimated)						
Synth Baseline	4.1	9	4,527	980	0.33	14.74
Synth CoT	4.1	9	4,527	980	0.33	14.74
Synth Mechanistic	4.1	9	4,527	980	0.33	14.74
GT Baseline	4.1	9	4,529	980	0.33	14.75
GT CoT	4.1	9	4,529	980	0.33	14.75
GT Mechanistic	4.1	9	4,529	980	0.33	14.75
RQ3 Deep Research (actual)						
Deep Research	4.1	3	291	240,913	67.70	197.00
Total						**2269.05**

Note. RQ1a costs from actual LLM Usage Stats. RQ1b estimated: correction + rejudging = 2× correctness tokens/edge. RQ3 from Deep Research logs. Models: 4.1 = GPT-4.1 ($2.00/$8.00 per 1M tokens), 5m = GPT-5-mini ($0.25/$2.00 per 1M tokens).
Reproduction: `python3 final_runs/cost_analysis/calculate_all_costs.py`

Table 26 reveals that Deep Research dominates per-edge cost (247K tokens/edge, \$0.87/edge) due to multi-agent orchestration and iterative search.

For a 66-edge CLD (Emergency Dept), generation plus correctness judging costs approximately \$0.29 and takes 6 minutes cumulative API time. Citation validation adds significant overhead (33 minutes, \$3.68 for GPT-4.1 or \$0.13 for GPT-5-mini). Linear $O(E)$ scaling enables predictable cost estimation for arbitrary CLD sizes.

16.3 Parallelization Speedup

We benchmarked parallelization using the configuration described in Methods: 34 edges (Depressive symptoms CLD), GPT-4.1 (temperature=0.3), worker counts of 1, 3, 5, and 10, with 3 runs per configuration for both correctness judging (short prompts) and citation judging (long prompts with retrieved citations).

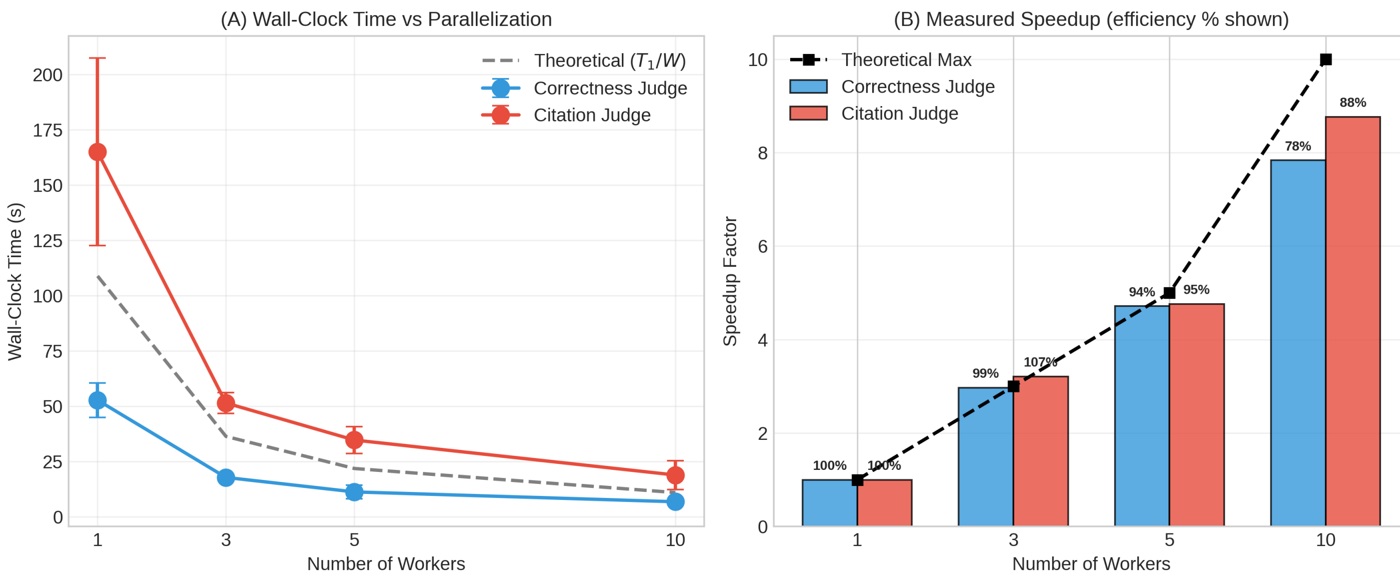

Figure 16: Parallelization benchmark (34 edges, GPT-4.1, 3 runs per config, mean ± SD). **(A)** Wall-clock time vs. worker count; dashed line shows theoretical perfect scaling (T_1/W). **(B)** Measured speedup with efficiency percentages.

Figure 16A shows citation judging (165s sequential) takes 3.1× longer than correctness (52.7s) due to longer prompts with retrieved citations. At 10 workers (Figure 16B), correctness achieves 7.8× speedup (78% efficiency) while citation achieves 8.8× speedup (88% efficiency). Citation judging shows slightly better parallel efficiency due to higher per-call latency dominating overhead.

Table 27: Parallelization Benchmark: Wall-Clock Time and Speedup by Judge Type

Workers	**Correctness Judge**			**Citation Judge**		
	Time (s)	Speedup	Eff.	Time (s)	Speedup	Eff.
1	52.7 ± 7.8	1.00×	100%	165.0 ± 42.4	1.00×	100%
3	17.7 ± 1.3	2.97×	99%	51.4 ± 4.6	3.21×	107%
5	11.2 ± 3.1	4.71×	94%	34.6 ± 6.1	4.76×	95%
10	6.7 ± 1.1	7.84×	78%	18.8 ± 6.5	8.77×	88%

Note. Model: GPT-4.1, temp=0.3. CLD: Depressive symptoms (34 edges). Time = wall-clock (mean ± SD, n=3). Speedup = T_1/T_W. Efficiency = Speedup/W × 100%.